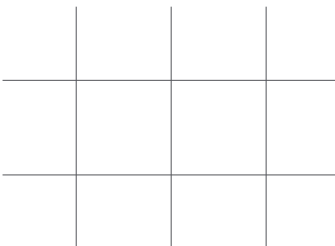


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PE 

civil: structural
practice exam

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About NCEES

NCEES is a nonprofit organization made up of the U.S. engineering and surveying licensing boards in all 50 states, the U.S. territories, and the District of Columbia. We develop and score the exams used for engineering and surveying licensure in the United States. NCEES also promotes professional mobility through its services for licensees and its member boards.

Engineering licensure in the United States is regulated by licensing boards in each state and territory. These boards set and maintain the standards that protect the public they serve. As a result, licensing requirements and procedures vary by jurisdiction, so stay in touch with your board (ncees.org/licensing-boards).

Exam format

Beginning in 2022, the PE Civil: Structural exam will be computer-based. It will contain 80 questions and be administered year-round via computer at approved Pearson VUE test centers. A 9-hour appointment time will include a tutorial, the exam, and a break. You will have 8 hours to complete the actual exam.

In addition to traditional multiple-choice questions with one correct answer, the exam will use common alternative item types such as

- Multiple correct options—allows multiple choices to be correct
- Point and click—requires examinees to click on part of a graphic to answer
- Drag and drop—requires examinees to click on and drag items to match, sort, rank, or label
- Fill in the blank—provides a space for examinees to enter a response to the question

To familiarize yourself with the format, style, and navigation of a computer-based exam, view the demos on ncees.org/ExamPrep and on the NCEES YouTube channel.

Examinee Guide

The *NCEES Examinee Guide* is the official guide to policies and procedures for all NCEES exams. During exam registration and again on exam day, examinees must agree to abide by the conditions in the *Examinee Guide*, which includes the CBT Examinee Rules and Agreement. You can download the *Examinee Guide* at ncees.org/exams. It is your responsibility to make sure you have the current version.

Scoring and reporting

Results for computer-based exams are typically available 7–10 days after you take the exam. You will receive an email notification from NCEES with instructions to view your results in your MyNCEES account. All results are reported as pass or fail.

Updates on exam content and procedures

Visit us at ncees.org/exams for updates on everything exam-related, including specifications, exam-day policies, scoring, and corrections to published exam preparation materials. This is also where you will register for the exam and find additional steps you should follow in your state to be approved for the exam.



EXAM SPECIFICATIONS

**NCEES Principles and Practice of Engineering Examination
CIVIL–STRUCTURAL CBT Exam Specifications
Effective Beginning in 2022**

- The exam topics have not changed since April 2015 when they were originally published.
- The exam is computer-based. It is closed book with electronic references. The NCEES *PE Civil Reference Handbook* and design standards applicable to the PE Civil–Structural exam are shown on ncees.org.
- Examinees have 9 hours to complete the exam, which contains 80 questions. The 9-hour time includes a tutorial and an optional scheduled break. Examinees work all questions.
- The exam uses both the International System of units (SI) and the US Customary System (USCS).
- The exam is developed with questions that will require a variety of approaches and methodologies, including design, analysis, and application.
- The examples specified in knowledge areas are not exclusive or exhaustive.

	Number of Questions
1. Project Planning	4–6
A. Quantity take-off methods	
B. Cost estimating	
C. Project schedules	
D. Activity identification and sequencing	
2. Means and Methods	3–5
A. Construction loads	
B. Construction methods	
C. Temporary structures and facilities	
3. Soil Mechanics	5–8
A. Lateral earth pressure	
B. Soil consolidation	
C. Effective and total stresses	
D. Bearing capacity	
E. Foundation settlement	
F. Slope stability	
4. Structural Mechanics	5–8
A. Dead and live loads	
B. Trusses	
C. Bending (e.g., moments and stresses)	
D. Shear (e.g., forces and stresses)	
E. Axial (e.g., forces and stresses)	
F. Combined stresses	
G. Deflection	
H. Beams	
I. Columns	
J. Slabs	
K. Footings	
L. Retaining walls	

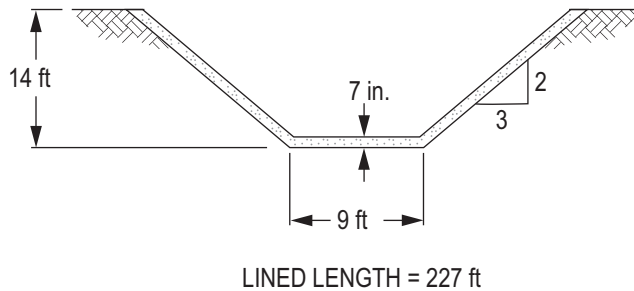
5. Hydraulics and Hydrology	6–9
A. Open-channel flow	
B. Stormwater collection and drainage (e.g., culvert, stormwater inlets, gutter flow, street flow, storm sewer pipes)	
C. Storm characteristics (e.g., storm frequency, rainfall measurement and distribution)	
D. Runoff analysis (e.g., Rational and SCS/NRCS methods, hydrographic application, runoff time of concentration)	
E. Detention/retention ponds	
F. Pressure conduit (e.g., single pipe, force mains, Hazen-Williams, Darcy-Weisbach, major and minor losses)	
G. Energy and/or continuity equation (e.g., Bernoulli)	
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B. Basic vertical curve elements	
C. Traffic volume (e.g., vehicle mix, flow, and speed)	
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A. Soil classification and boring log interpretation	
B. Soil properties (e.g., strength, permeability, compressibility, phase relationships)	
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C. Temporary and permanent soil erosion and sediment control (e.g., construction erosion control and permits, sediment transport, channel/outlet protection)	
D. Impact of construction on adjacent facilities	
E. Safety (e.g., construction, roadside, work zone)	
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2. Live loads	
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5. Seismic loads	
6. Moving loads (e.g., vehicular, cranes)	
7. Snow, rain, ice	
8. Impact loads	
9. Earth pressure and surcharge loads	
10. Load paths (e.g., lateral and vertical)	
11. Load combinations	
12. Tributary areas	

B. Forces and load effects	9–14
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2. Axial (e.g., tension and compression)	
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4. Flexure	
5. Deflection	
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10. Design and Details of Structures	16–24
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2. Steel (e.g., structural, reinforcing, cold-formed)	
3. Timber	
4. Masonry (e.g., brick veneer, CMU)	
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1. Horizontal members (e.g., beams, slabs, diaphragms)	
2. Vertical members (e.g., columns, bearing walls, shear walls)	
3. Systems (e.g., trusses, braces, frames, composite construction)	
4. Connections (e.g., bearing, bolted, welded, embedded, anchored)	
5. Foundations (e.g., retaining walls, footings, combined footings, slabs, mats, piers, piles, caissons, drilled shafts)	
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2. American Concrete Institute (ACI 318, 530)	
3. Precast/Prestressed Concrete Institute (PCI Design Handbook)	
4. Steel Construction Manual (AISC)	
5. National Design Specification for Wood Construction (NDS)	
6. LRFD Bridge Design Specifications (AASHTO)	
7. Minimum Design Loads for Buildings and Other Structures (ASCE 7)	
8. American Welding Society (AWS D1.1, D1.2, and D1.4)	
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2. Submittals	
3. Formwork	
4. Falsework and scaffolding	
5. Shoring and reshoring	
6. Concrete maturity and early strength evaluation	
7. Bracing	
8. Anchorage	
9. OSHA regulations	
10. Safety management	

PE CIVIL: STRUCTURAL PRACTICE EXAM

PE CIVIL: STRUCTURAL PRACTICE EXAM

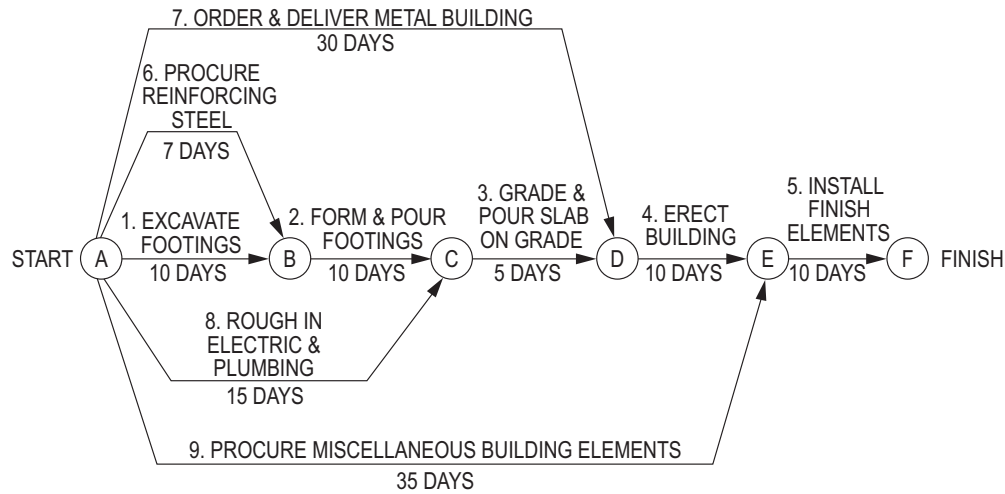
1. A 227-ft length of canal is to be lined with concrete for erosion control. With 12% allowance for waste and overexcavation, the volume (yd^3) of concrete that must be delivered is most nearly:



- A. 234
- B. 280
- C. 292
- D. 327
2. A track loader has an initial cost of \$75,000 and a salvage value of \$10,000 at the end of its expected life of 10 years. Based on the straight-line method of depreciation, at the end of the track loader's 8th year, its book value is most nearly:
- A. \$10,000
- B. \$15,000
- C. \$23,000
- D. \$48,750
3. The budgeted labor amount for an excavation task is \$4,000. The hourly labor cost is \$50 per worker, and the workday is 8 hours. Two workers are assigned to excavate the material. The time (days) available for the workers to complete this task is most nearly:
- A. 3
- B. 4
- C. 5
- D. 12.5

PE CIVIL: STRUCTURAL PRACTICE EXAM

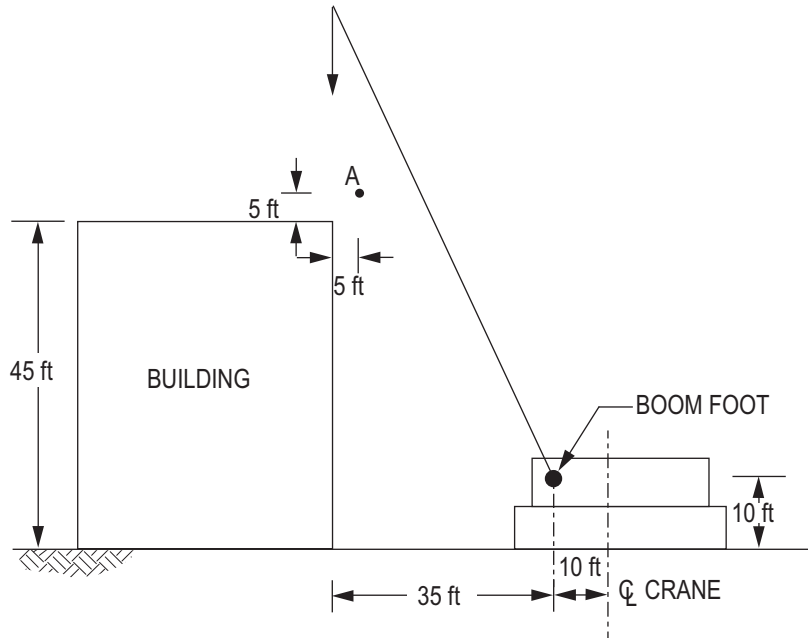
4. A CPM arrow diagram is shown below. Nine activities have been estimated with durations ranging from 5 to 35 days. The minimum time (days) required to finish the project is most nearly:



- A. 40
- B. 42
- C. 45
- D. 50
5. A bridge is to be jacked up to replace its bearings. The design requires a hydraulic ram with a minimum capacity of 1,000 kilonewtons (kN). The hydraulic rams that are available are rated in tons (2,000 lb/ton). The **minimum** size (tons) ram to use is most nearly:
- A. 1,110
- B. 250
- C. 150
- D. 100

PE CIVIL: STRUCTURAL PRACTICE EXAM

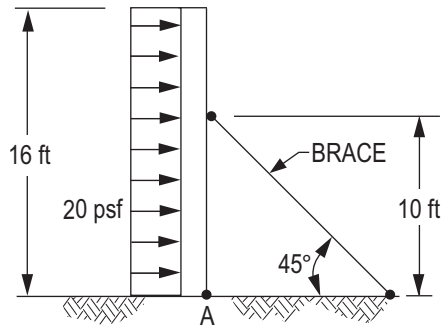
6. A crane with a 100-ft boom is being used to set a small load on the roof of the building shown. The minimum standoff (Point A) from the corner of the building to the centerline of the boom is indicated. What is the maximum distance (ft) from the edge of the building that the load can be placed on the roof?



- ☐ A. 16
- ☐ B. 25
- ☐ C. 30
- ☐ D. 36

PE CIVIL: STRUCTURAL PRACTICE EXAM

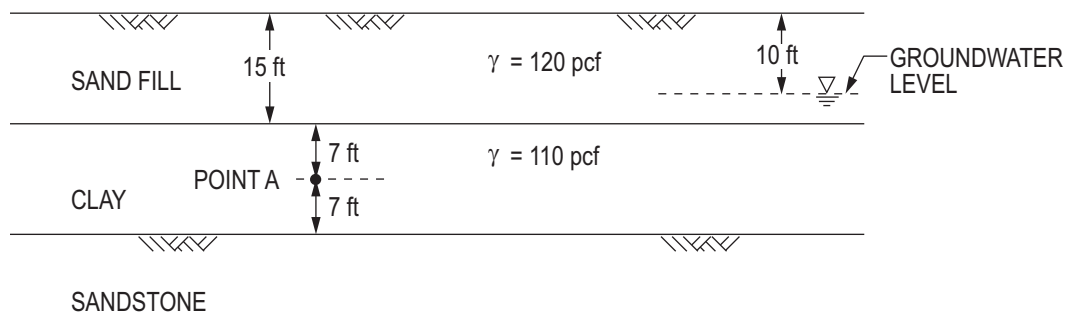
7. A wall form subjected to a wind load of 20 psf is prevented from overturning by diagonal braces spaced at 8 ft on center along the length of the wall form as shown in the figure. The connection at the base of the form at Point A is equivalent to a hinge. Ignore the weight of the form. The axial force (lb) resisted by the brace is most nearly:



- ☐ A. 2,050
 - ☐ B. 2,560
 - ☐ C. 2,900
 - ☐ D. 4,525
8. Which one of the following statements regarding lateral earth pressures is correct?
- ☐ A. The lateral strain required to fully mobilize the soil passive pressure is considerably smaller than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ B. The lateral strain required to fully mobilize the soil passive pressure is slightly smaller than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ C. The lateral strain required to fully mobilize the soil passive pressure is slightly larger than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ D. The lateral strain required to fully mobilize the soil passive pressure is considerably larger than the lateral strain required to fully mobilize the soil active pressure.

PE CIVIL: STRUCTURAL PRACTICE EXAM

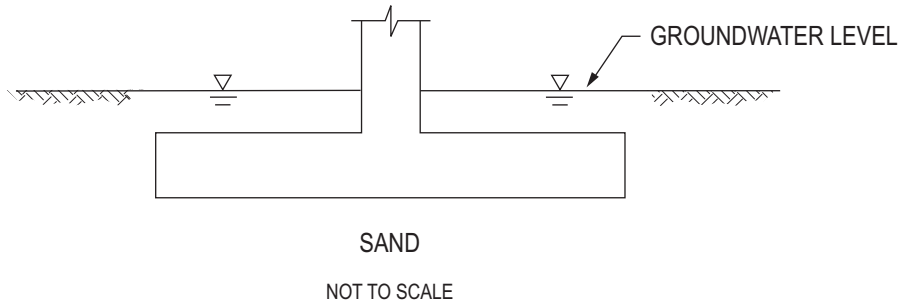
9. Site preparation and grading require the placement of 20 ft of new fill. An analysis of the resulting consolidation of the underlying soft, saturated, compressible deposits reveals a mean consolidation settlement of 22 in. affecting a 21.5-acre area. Prefabricated wick drains will be used to accelerate the settlement to meet the project schedule. Because of contamination from the former site use, the effluent from the wick drains must be collected and treated prior to disposal at an estimated cost of \$0.25 per gallon. Assume no loss of effluent during collection. The estimated treatment and disposal cost for the wick drain effluent at this site is most nearly:
- A. \$430,000
 - B. \$3,200,000
 - C. \$5,200,000
 - D. \$35,000,000
10. A soil profile is shown in the figure. The effective vertical stress (psf) at Point A is most nearly:



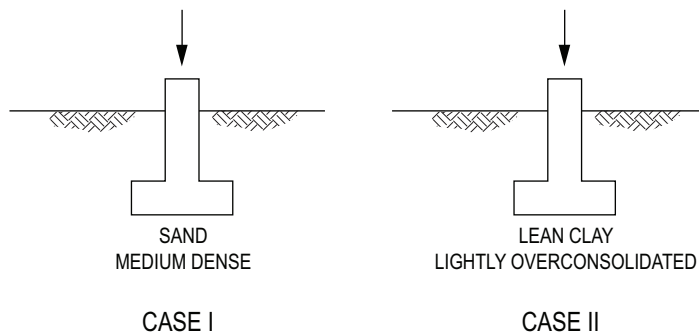
- A. 1,270
- B. 1,820
- C. 2,140
- D. 2,570

PE CIVIL: STRUCTURAL PRACTICE EXAM

11. A bridge footing is to be constructed in sand. The groundwater level is at the ground surface. The ultimate bearing capacity is based on what type of soil unit weight?



- ☐ A. Buoyant unit weight
 - ☐ B. Saturated unit weight
 - ☐ C. Dry unit weight
 - ☐ D. Total unit weight
12. The figure shows two identical building footings with the same load but constructed in two different soil types. Which of the following statements is most correct?



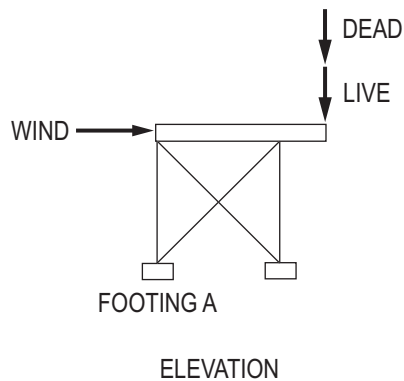
- ☐ A. The long-term settlement for Case I is less than Case II.
- ☐ B. The long-term settlement for Case II is less than Case I.
- ☐ C. The long-term settlements are the same for both cases.
- ☐ D. Settlement is not a concern for either case.

PE CIVIL: STRUCTURAL PRACTICE EXAM

13. The minimum factor of safety against rotational failure for permanent slopes under long-term, nonseismic conditions influencing occupied structures is closest to:

- ☐ A. 1.0
- ☐ B. 1.1
- ☐ C. 1.5
- ☐ D. 3.0

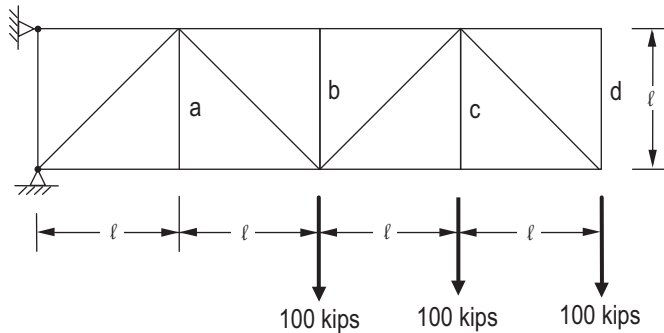
14. In the figure below, what load combination produces the maximum uplift on Footing A?



- ☐ A. Dead
- ☐ B. Dead + live
- ☐ C. Dead + wind
- ☐ D. Dead + live + wind

PE CIVIL: STRUCTURAL PRACTICE EXAM

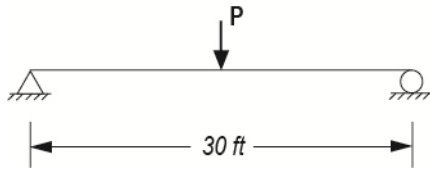
15. A simply supported truss is loaded as shown in the figure. The loads (kips) for Members b and c are most nearly:



- ☐ A. Member b = 0
Member c = 0
 - ☐ B. Member b = 0
Member c = 100
 - ☐ C. Member b = 100
Member c = 0
 - ☐ D. Member b = 100
Member c = 100
16. Consider two beams with equal cross sections, made of the same material, having the same support conditions, and each loaded with equal uniform load per length. One beam is twice as long as the other. The maximum bending stress in the longer beam is larger by a factor of:
- ☐ A. 1.25
 - ☐ B. 2
 - ☐ C. 3
 - ☐ D. 4

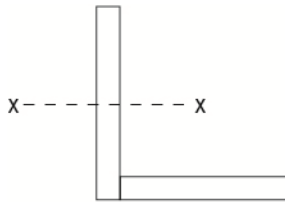
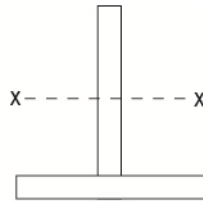
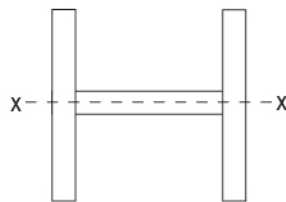
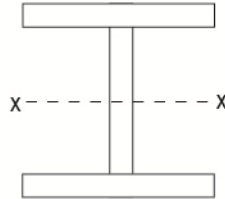
PE CIVIL: STRUCTURAL PRACTICE EXAM

17. The point load (kips) placed at the centerline of a 30-ft beam that produces the same maximum shear in the beam as a uniform load of 1 kip/ft is most nearly:



- ☐ A. 7.5
- ☐ B. 15
- ☐ C. 30
- ☐ D. 60

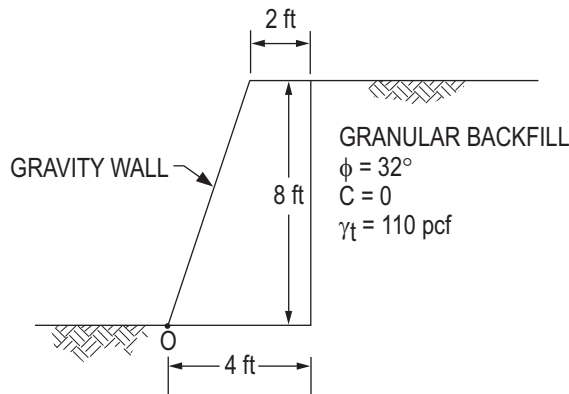
18. The beam sections shown are fabricated from 1/2-in. $\times$ 6-in. steel plates. Which of the following cross sections will provide the greatest flexural rigidity about the x-axis?

**Option A****Option B****Option C****Option D**

- ☐ A. Option A
- ☐ B. Option B
- ☐ C. Option C
- ☐ D. Option D

PE CIVIL: STRUCTURAL PRACTICE EXAM

19. A concrete gravity retaining wall having a unit weight of 150 pcf is shown in the figure. Use the Rankine active earth pressure theory and neglect wall friction. The factor of safety against overturning about the toe at Point O is most nearly:



- A. 3.1
- B. 2.5
- C. 2.2
- D. 0.3
20. A drainage basin produces a stormwater runoff volume of 25.0 acre-ft, which must be drained through a rectangular channel that is 4 ft wide and 2 ft deep and has a uniform slope of 0.2%. Assume a Manning roughness coefficient of 0.022 and a constant depth of flow of 1.5 ft. The time (hours) it will take to discharge the runoff is most nearly:
- A. 12.5
- B. 16.4
- C. 18.5
- D. 25.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

21. Two identical 12-in. storm sewers flow full at a 2% slope into a junction box. A single larger pipe of the same material and slope flows out of the box. Assume the following pipe sizes are commercially available. The minimum size (in.) of this downstream pipe designed to flow full is most nearly:
- ☐ A. 16
 - ☐ B. 18
 - ☐ C. 20
 - ☐ D. 24
22. The following table represents the rainfall recorded from all rain gauges located in and around a drainage area:

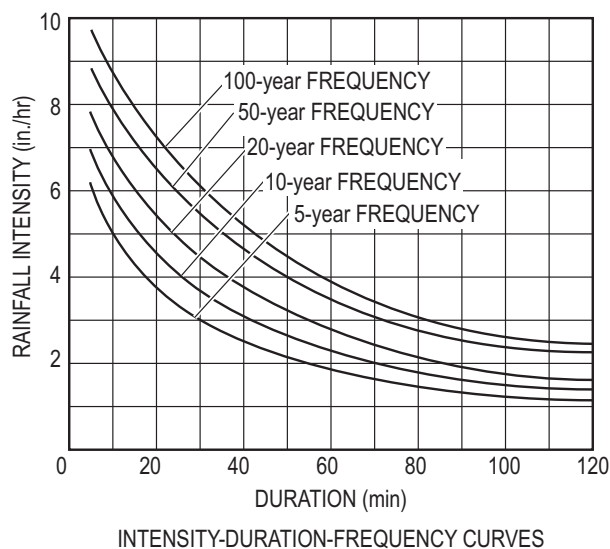
Gauge	A	B	C	D	E	F	G	H	I	J	K
Rainfall (in.)	2.1	3.6	1.3	1.5	2.6	6.1	5.1	4.8	4.1	2.8	3.0

Use the arithmetic mean method. The average precipitation (in.) for the drainage area is most nearly:

- ☐ A. 3.4
- ☐ B. 3.7
- ☐ C. 4.1
- ☐ D. 37.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

23. The rational method must be used to determine the maximum runoff rate for a 90-acre downtown area. The time of concentration for the 50-year frequency storm is 1 hour. Intensity-duration-frequency curves and a table of runoff coefficients are provided. The maximum runoff rate (cfs), based on the maximum runoff coefficient for a 50-year storm, is most nearly:

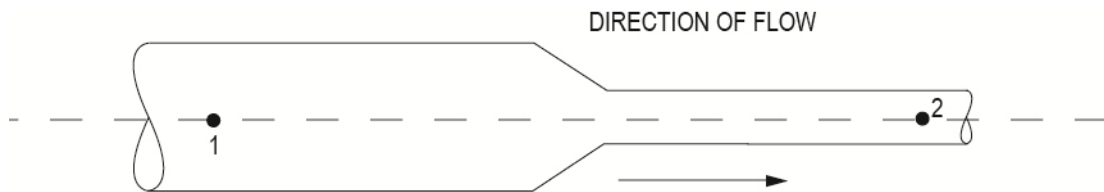


- A. 160
 ○ B. 220
 ○ C. 300
 ○ D. 340

Description of Area	Runoff Coefficients
Business	
Downtown areas	0.70–0.95
Neighborhood areas	0.50–0.70
Residential	
Single-family areas	0.30–0.50
Multiunits, detached	0.40–0.60
Multiunits, attached	0.60–0.75
Residential (suburban)	0.25–0.40
Apartment dwelling areas	0.50–0.70
Industrial	
Light areas	0.50–0.80
Heavy areas	0.60–0.90
Parks, cemeteries	0.10–0.25
Playgrounds	0.20–0.35
Railroad yard areas	0.20–0.40
Unimproved areas	0.10–0.30
Streets	
Asphalt	0.70–0.95
Concrete	0.80–0.95
Brick	0.70–0.85
Drives and walks	0.75–0.85

PE CIVIL: STRUCTURAL PRACTICE EXAM

24. A stormwater drainage ditch with a maximum capacity of 10 cfs discharges into a detention basin. The detention basin volume is 400,000 gal. During a storm event, the average discharge into the detention basin was 1.5 cfs. The time (hours) to fill the empty basin is most nearly:
- ☐ A. 1.5
 - ☐ B. 9.9
 - ☐ C. 11.1
 - ☐ D. 74.1
25. Assume fully turbulent flow in a 1,650-ft section of 3-ft-diameter pipe. The Darcy–Weisbach friction factor f is 0.0115. There is a 5-ft drop in the energy grade line over the section. The flow rate (cfs) is most nearly:
- ☐ A. 16
 - ☐ B. 29
 - ☐ C. 50
 - ☐ D. 810
26. Assume that Bernoulli's equation applies (ignore head losses) to the pipe flow shown in the figure. Which of the following statements is most correct?



- ☐ A. Pressure head increases from 1 to 2.
- ☐ B. Pressure head decreases from 1 to 2.
- ☐ C. Pressure head remains unchanged from 1 to 2.
- ☐ D. Bernoulli's equation does not include pressure head.

PE CIVIL: STRUCTURAL PRACTICE EXAM

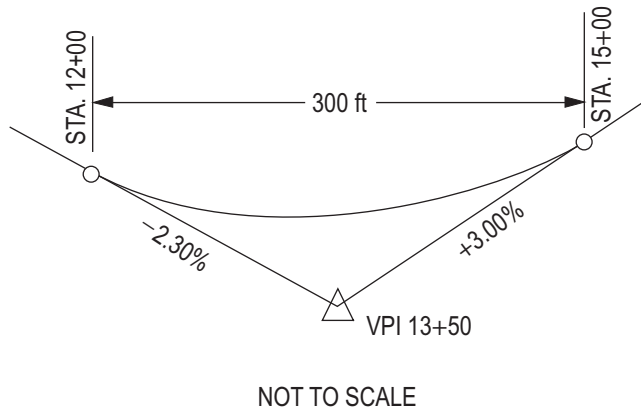
27. The following information is for a proposed horizontal curve in a new subdivision:

PI station	12+40.00
Degree of curve	10°
Deflection angle	12°30'

The station of the PT is most nearly:

- ☐ A. 12+79.80
- ☐ B. 12+80.10
- ☐ C. 13+02.00
- ☐ D. 13+64.75

28. For the sag vertical curve shown, the tangent slope at Station 14+00 is most nearly:



- ☐ A. +0.53%
- ☐ B. +1.23%
- ☐ C. +2.12%
- ☐ D. +2.77%

PE CIVIL: STRUCTURAL PRACTICE EXAM

29. An interstate highway has the following traffic count data for a day in each month as shown below:

Jan.	63,500
Feb.	62,100
Mar.	64,400
Apr.	64,900
May	75,800
June	77,300
July	78,950
Aug.	77,200
Sept.	70,050
Oct.	69,000
Nov.	66,000
Dec.	<u>64,000</u>
Annual Total	833,200

To determine the seasonal average daily traffic volume for June through August, the seasonal factor to be multiplied by the average daily traffic volume is most nearly:

- ☐ A. 0.28
 - ☐ B. 0.89
 - ☐ C. 1.02
 - ☐ D. 1.12
30. The most essential criteria for proper soil classification using the Unified Soil Classification System or the AASHTO Soil Classification System are:
- ☐ A. water content and soil density
 - ☐ B. Atterberg limits and specific gravity
 - ☐ C. grain-size distribution and water content
 - ☐ D. grain-size distribution and Atterberg limits

PE CIVIL: STRUCTURAL PRACTICE EXAM

31. The standard penetration test (SPT) is widely used as a simple and economic means of obtaining which of the following?
- ☐ A. A measurement of soil compressibility expressed in terms of a compression index
 - ☐ B. A direct measurement of the undrained shear strength
 - ☐ C. An indirect indication of the relative density of cohesionless soils
 - ☐ D. A direct measurement of the angle of internal friction
32. A department of transportation must remove and replace a 12-ft $\times$ 20-ft concrete slab on an interstate facility. To minimize disruption to traffic, the work must be completed during an 8-hour nighttime work shift. Nighttime temperatures average 50°F. If the minimum required compressive strength is 3,500 psi, the concrete mix most likely consists of:
- ☐ A. coarse aggregate, sand, Type II cement, chemical accelerator
 - ☐ B. sand, Type III cement, water, chemical accelerator
 - ☐ C. coarse aggregate, sand, Type V cement, water, chemical accelerator
 - ☐ D. coarse aggregate, sand, Type III cement, water, chemical accelerator

PE CIVIL: STRUCTURAL PRACTICE EXAM

33. Fatigue in steel can be the result of:

- ☐ A. a reduction in strength due to cyclical loads
- ☐ B. deformation under impact loads
- ☐ C. deflection due to overload
- ☐ D. expansion due to corrosion

34. Sample concrete cylinders that are 6 inches in diameter and 12 inches high are tested to determine the compressive strength of the concrete f'_c . The test results are as follows:

Sample	Axial Compressive Failure Load (lb)
1	65,447
2	63,617
3	69,872

Based on the test results, the average 28-day compressive strength (psi) is most nearly:

- ☐ A. 615
- ☐ B. 2,250
- ☐ C. 2,340
- ☐ D. 2,470

PE CIVIL: STRUCTURAL PRACTICE EXAM

35. During testing of a sample in the laboratory, the following soil data are collected:

Combined weight of compacted soil sample and mold = 9.11 lb

Water content of soil sample = 11.5%

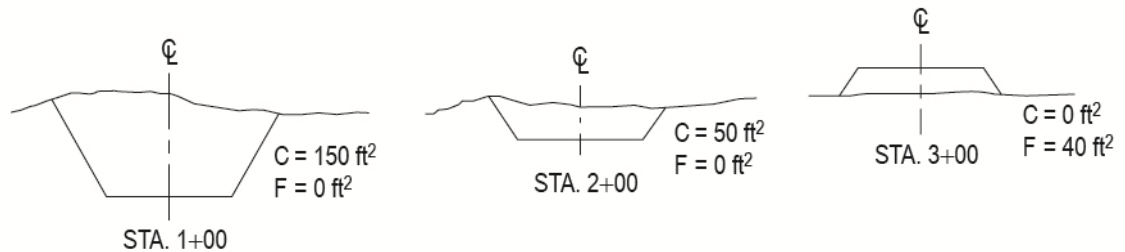
Weight of mold = 4.41 lb

Volume of mold = 0.03 ft^3

The dry unit weight (pcf) of the soil sample is most nearly:

- ☐ A. 160
- ☐ B. 140
- ☐ C. 127
- ☐ D. 125

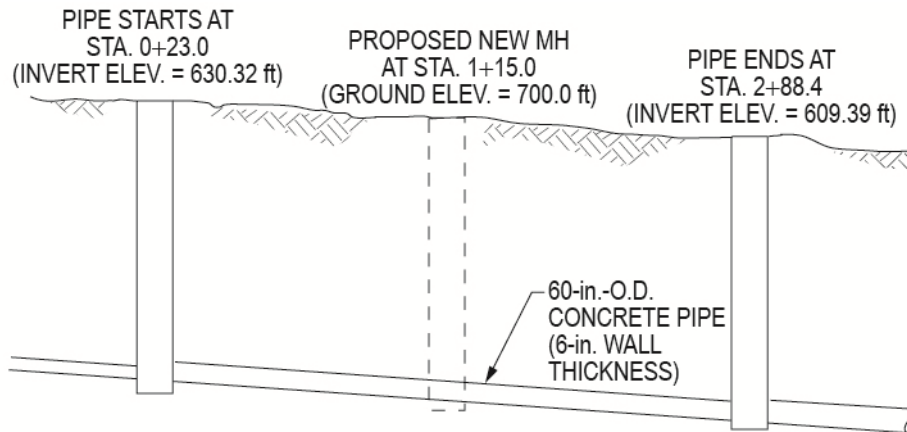
36. Refer to the figure. The net excess excavated material (yd^3) from Station 1+00 to Station 3+00 is most nearly:



- ☐ A. 160
- ☐ B. 262
- ☐ C. 390
- ☐ D. 463

PE CIVIL: STRUCTURAL PRACTICE EXAM

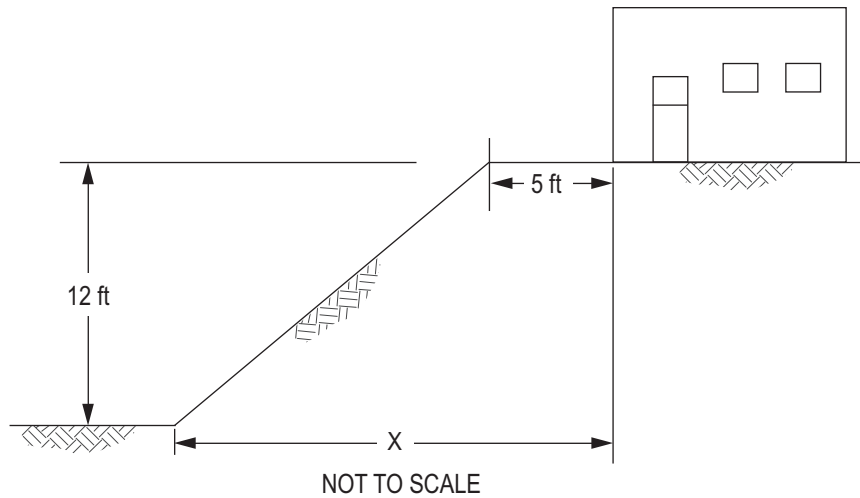
37. An existing pipe connects two maintenance holes (MH). A third MH is planned between the two. At the new MH, the elevation (ft) of the top of the pipe is most nearly:



- ☐ A. 623.06
 - ☐ B. 627.56
 - ☐ C. 628.06
 - ☐ D. 628.56
38. Which of the following is **not** a stormwater erosion classification?
- ☐ A. Sheet erosion
 - ☐ B. Rill erosion
 - ☐ C. Gully erosion
 - ☐ D. Rushing erosion

PE CIVIL: STRUCTURAL PRACTICE EXAM

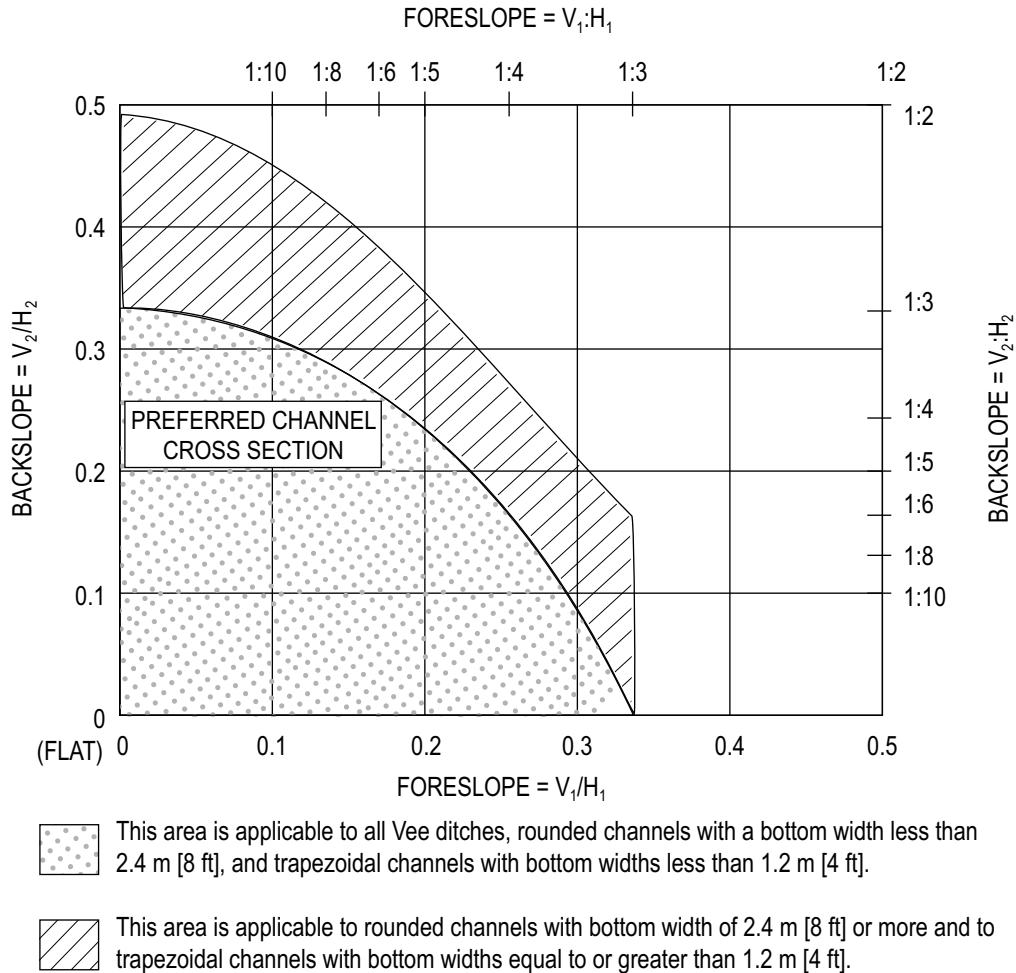
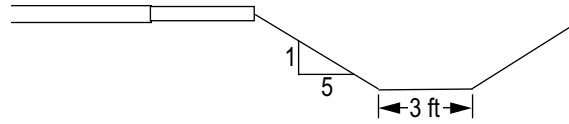
39. Based on the soil classification system found in the federal OSHA 29 CFR 1926, Subpart P, Excavations, the soil adjacent to an existing building has been classified as Type B. An undisturbed perimeter strip that is 5 ft wide is to be maintained along the face of the building. The excavation is to be 12 ft deep. To meet OSHA excavation requirements, the minimum horizontal distance X (ft) from the toe of the slope to the face of the structure is most nearly:



- ☐ A. 11
- ☐ B. 14
- ☐ C. 17
- ☐ D. 23

PE CIVIL: STRUCTURAL PRACTICE EXAM

40. Based on the criteria provided, the steepest backslope (H:V) preferred in the ditch shown is most nearly:



Adapted from AASHTO *Roadside Design Guide*, 4th edition, 2011.

- ☐ A. 2:1
- ☐ B. 3:1
- ☐ C. 5:1
- ☐ D. 6:1

PE CIVIL: STRUCTURAL PRACTICE EXAM

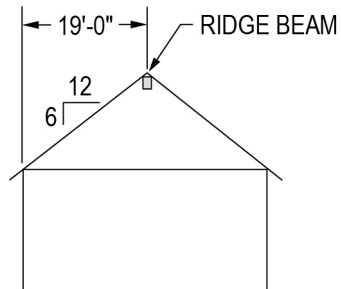
41. The following data apply to the structure shown in the figure:

Ground snow load = 20 psf
Roof is fully exposed with wood shingles
Open terrain
Risk Category I
Unheated structure

Design Code:

ASCE 7: *Minimum Design Loads for Buildings and Other Structures*, 3rd printing, 2010.

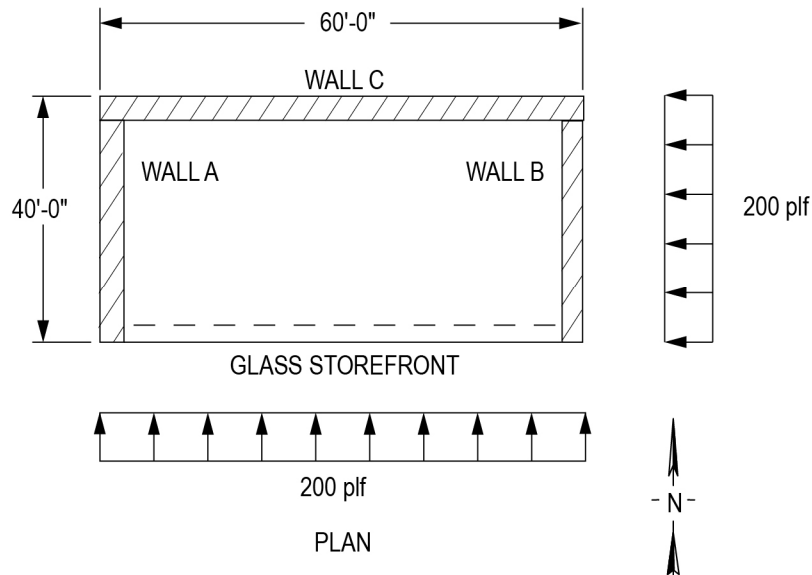
Roof joists span from the exterior walls to the ridge beam. The design snow load (psf) for roof joists is most nearly:



- ☐ A. 10
- ☐ B. 12
- ☐ C. 16
- ☐ D. 20

PE CIVIL: STRUCTURAL PRACTICE EXAM

42. The figure shows a one-story building with CMU walls on three sides and a glass storefront. The uniform load shown in the figure is the wind reaction acting at the top of the wall, directly into the diaphragm. The resultant force (kips) on Wall B from the wind acting on the front wall is most nearly:

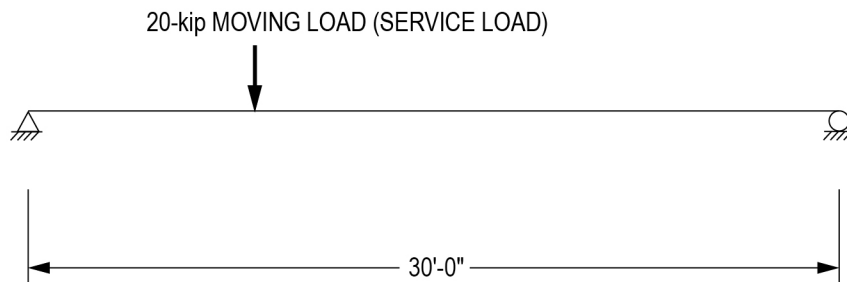


- ☐ A. 4
- ☐ B. 6
- ☐ C. 8
- ☐ D. 12

PE CIVIL: STRUCTURAL PRACTICE EXAM

43. A two-story 100-ft $\times$ 100-ft industrial facility in Seismic Design Category A has a second floor that is used for storage and a 200-psf reduced floor live load. If the facility has a flat roof that experiences a snow load of 50 psf and the total dead load of the building is 200,000 lb, the minimum lateral force E (lb) for the entire building is most nearly:
- ☐ A. 2,000
 - ☐ B. 7,000
 - ☐ C. 22,000
 - ☐ D. 27,000
44. A moving load is applied to a beam as shown in the figure. The maximum service moment (ft-kips) carried by the beam is _____.

Enter your response in the blank.



PE CIVIL: STRUCTURAL PRACTICE EXAM

45. A building floor is supported by steel beams as shown in the figure. The floor beams are spaced 10 ft o.c. at either side of the beam shown.

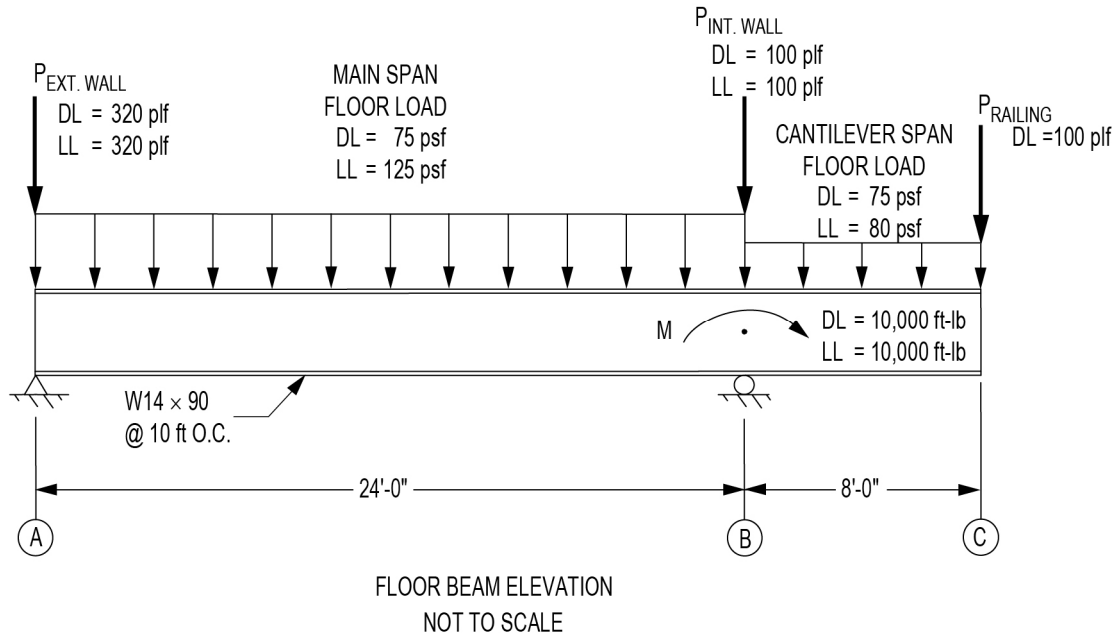
Assumptions:

All loads act simultaneously.

Neglect beam self-weight.

The live loads are non-reducible.

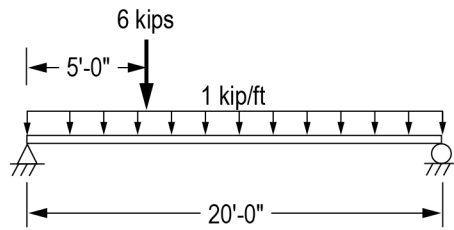
The total service load reaction (kips) at the support at A is most nearly:



- ☐ A. 3
- ☐ B. 14
- ☐ C. 28
- ☐ D. 35

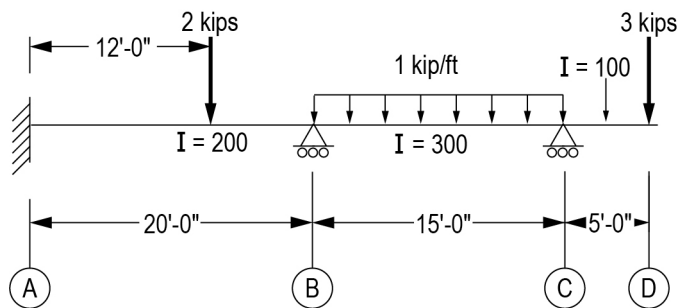
PE CIVIL: STRUCTURAL PRACTICE EXAM

46. The beam shown in the figure is loaded with a uniform load of 1 kip/ft and a point load of 6 kips located 5 ft from the left end. The location of the maximum moment (ft) measured from the right end is most nearly:



- ☐ A. 5.0
- ☐ B. 8.5
- ☐ C. 10.0
- ☐ D. 11.5

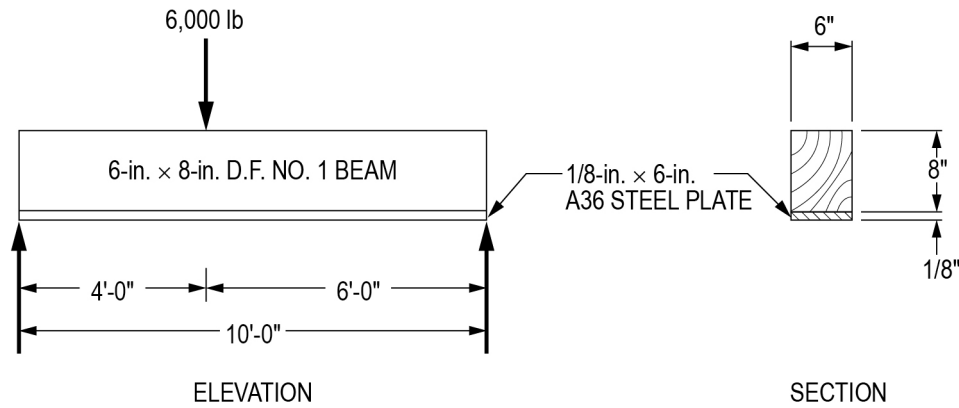
47. A loaded beam is shown in the figure. Using the method of moment distribution, the unbalanced portion of the fixed-end moments (ft-kips) at Joint B is most nearly:



- ☐ A. 19
- ☐ B. 13
- ☐ C. 6
- ☐ D. 0

PE CIVIL: STRUCTURAL PRACTICE EXAM

48. The beam shown in the figure is simply supported, and the dead load is negligible. A steel plate is attached to the beam sufficiently to fully develop composite section. $E_{\text{wood}} = 1.6 \times 10^6$ psi and $E_{\text{steel}} = 29 \times 10^6$ psi. The tensile stress in the steel (ksi) is most nearly:



- ☐ A. 2.6
- ☐ B. 14.4
- ☐ C. 18.6
- ☐ D. 23.5

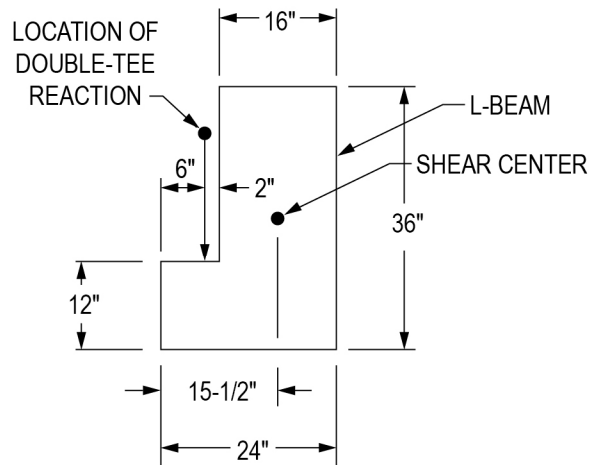
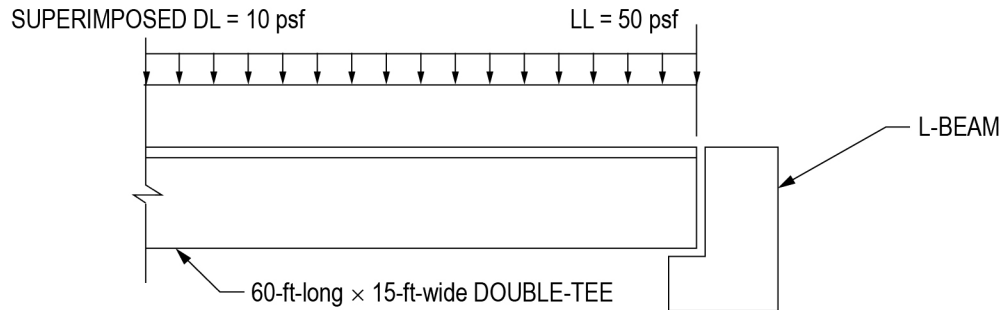
PE CIVIL: STRUCTURAL PRACTICE EXAM

49. A double-tee supported by an L-beam is loaded as shown in the figure.

Design Code:

ACI 318: *Building Code Requirements for Structural Concrete*, 2014.

The magnitude of the ultimate torsion (ft-kips) induced into the L-beam, by superimposed dead and live load applied on the double-tee, is most nearly:



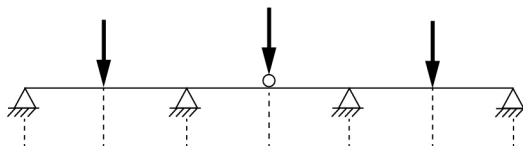
DETAIL

NOT TO SCALE

- ☐ A. 18.7
- ☐ B. 21.4
- ☐ C. 28.6
- ☐ D. 32.8

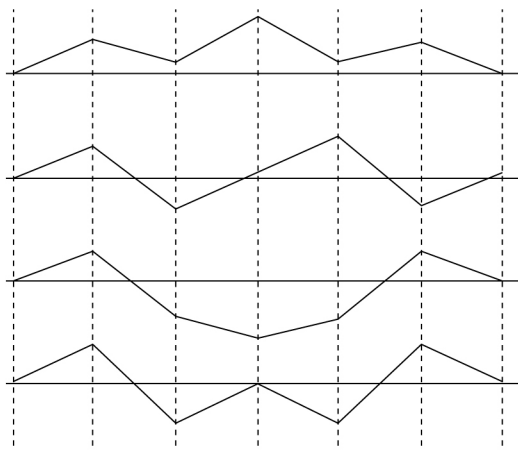
PE CIVIL: STRUCTURAL PRACTICE EXAM

50. A beam is loaded as shown:

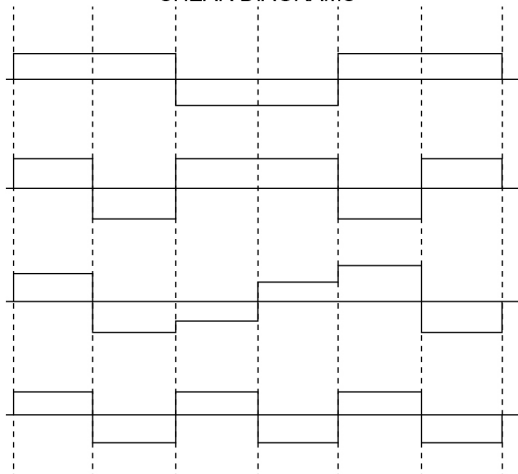


Select the two diagrams that most accurately represent the shear and bending moment for the loaded beam shown.

MOMENT DIAGRAMS

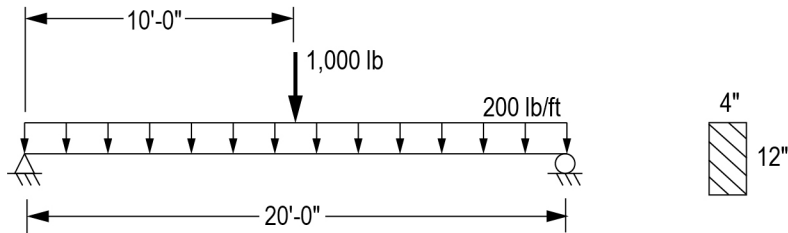


SHEAR DIAGRAMS



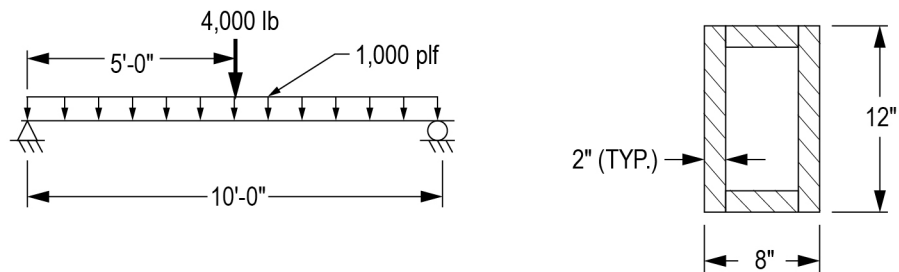
PE CIVIL: STRUCTURAL PRACTICE EXAM

51. Referring to the figure, the maximum flexural stress (psi) in the member is most nearly:



- ☐ A. 5,625
- ☐ B. 3,125
- ☐ C. 1,875
- ☐ D. 1,565

52. The figure shows a section with elements that are adequately connected to ensure composite action. Assuming $Q = 85 \text{ in}^3$, the maximum horizontal shear stress (psi) in the member is most nearly:



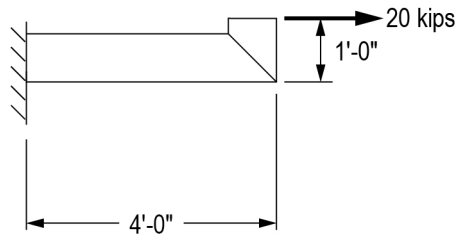
- ☐ A. 90
- ☐ B. 150
- ☐ C. 300
- ☐ D. 400

PE CIVIL: STRUCTURAL PRACTICE EXAM

53. The pipe member shown in the figure has a constant section and the following properties:

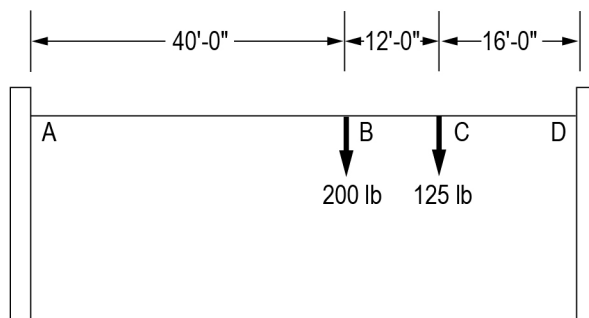
Outside diameter	10 in.
Moment of inertia	90 in^4
Area	7 in^2

Neglecting the weight of the pipe, the maximum compressive stress (ksi) at the support is most nearly:



- ☐ A. 2.9
- ☐ B. 4.9
- ☐ C. 13.3
- ☐ D. 16.2

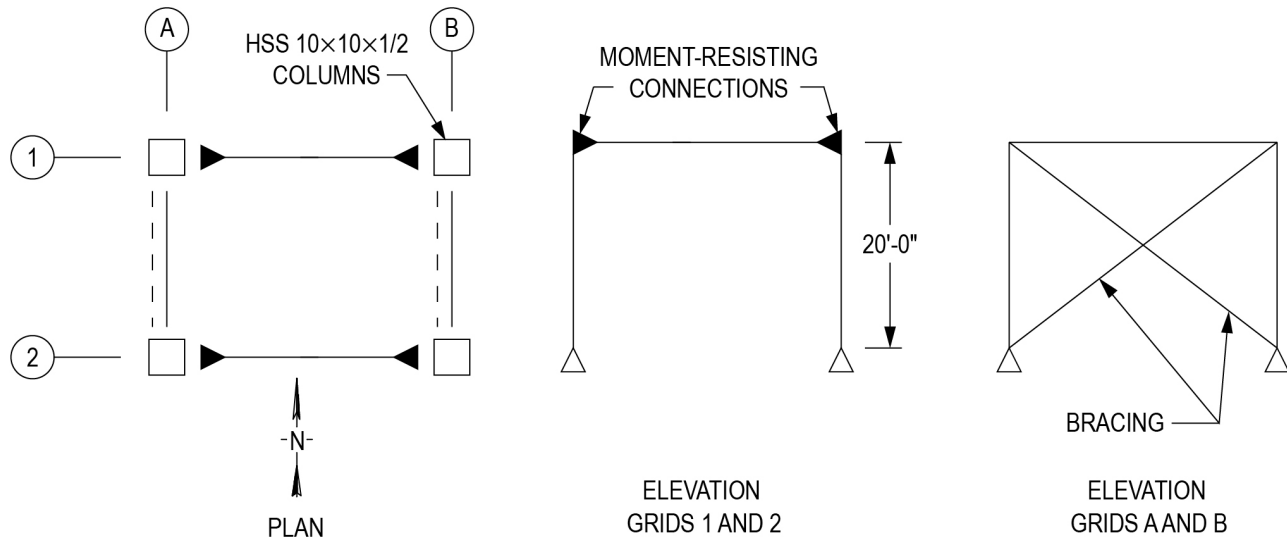
54. A cable carrying traffic signal loads is shown in the figure. The deflection at point B is 12 in. Neglect elongation of the cable. The tension force (lb) in the cable between C and D is most nearly:



- ☐ A. 4,685
- ☐ B. 4,476
- ☐ C. 1,816
- ☐ D. 213

PE CIVIL: STRUCTURAL PRACTICE EXAM

55. A structure is shown in the figure. For a stress check of column A1, which of the following statements is true?



- ☐ A. Buckling in the north-south direction will control the column capacity.
- ☐ B. Buckling in the east-west direction will control the column capacity.
- ☐ C. Buckling in either direction might control the column capacity.
- ☐ D. Buckling will not control since both directions are stable.

PE CIVIL: STRUCTURAL PRACTICE EXAM

56. Normal-weight concrete with a maximum aggregate size of 1 1/2 in., $f'_c = 4,000$ psi, has moderate exposure to freezing and thawing. Neglecting tolerance, the recommended minimum air content is most nearly:
- ☐ A. 4.5%
 - ☐ B. 5.0%
 - ☐ C. 5.5%
 - ☐ D. 6.0%
57. Which of the following characterizes the typical failure of an over-reinforced concrete beam that has a reinforcement ratio higher than the balanced ratio?
- ☐ A. The steel yields, and large deflections and tensile cracks will be observed prior to failure.
 - ☐ B. The concrete crushes, and large deflections and tensile cracks will be observed prior to failure.
 - ☐ C. The steel yields, and the beam fails suddenly without warning.
 - ☐ D. The concrete crushes, and the beam fails suddenly without warning.

PE CIVIL: STRUCTURAL PRACTICE EXAM

58. The figure shows a wall construction plan.

Design Code:

NDS: *National Design Specification for Wood Construction ASD/LRFD*, 2015 edition and
National Design Specification Supplement, Design Values for Wood Construction, 2015 edition.

Material:

Douglas Fir-Larch No. 2

Assumptions:

The gypsum board provides stud weak axis bracing.

$$C_D = 1.6$$

$$C_M = 1.0$$

$$C_t = 1.0$$

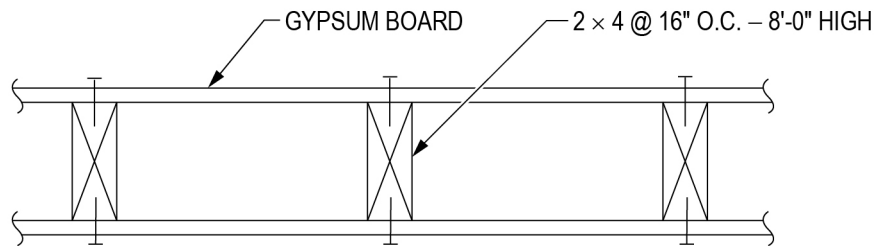
$$C_F = 1.15$$

$$C_i = 1.00$$

$$C_p = 0.33 \text{ strong axis}$$

$$C_p = 1.0 \text{ weak axis}$$

Using allowable stress design, the maximum vertical load (lb) per stud is most nearly:



- ☐ A. 2,700
- ☐ B. 3,300
- ☐ C. 3,700
- ☐ D. 4,300

PE CIVIL: STRUCTURAL PRACTICE EXAM

59. The following information applies to the masonry beam section shown:

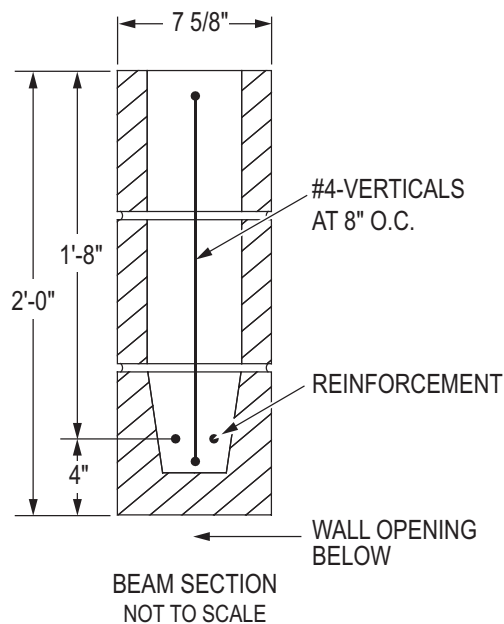
$$f'_m = 1,500 \text{ psi}$$

$$f_y = 60,000 \text{ psi}$$

Design Code:

TMS 402/602: *Building Code Requirements and Specification for Masonry Structures* (and related commentaries), 2013.

Use allowable stress design. Assume shear load is resisted by shear reinforcement only. The maximum allowable shear force (kips) for the beam is most nearly:



- ☐ A. 10
- ☐ B. 12
- ☐ C. 14
- ☐ D. 18

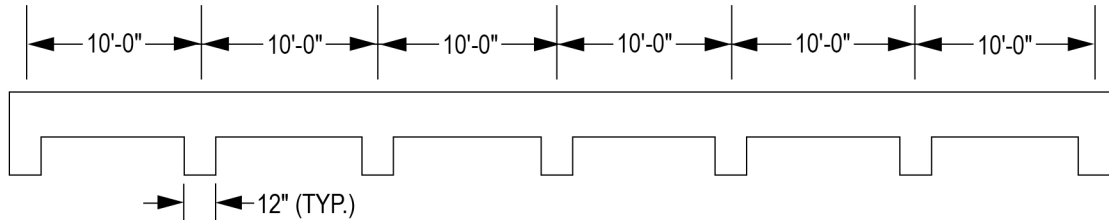
PE CIVIL: STRUCTURAL PRACTICE EXAM

60. A concrete slab of uniform thickness is continuous for six spans and is supported by 12-in.-wide concrete beams at 10 ft o.c. The following loads are applied to the slab:

DL 100 psf (includes self-weight)

LL 50 psf

Using ACI 318-14's alternate to frame analysis, the maximum factored negative moment in the slab (ft-kips/ft) is most nearly:



- ☐ A. 1.35
- ☐ B. 1.47
- ☐ C. 1.67
- ☐ D. 1.80

PE CIVIL: STRUCTURAL PRACTICE EXAM

61. The W10×22 steel beam ($F_y = 50$ ksi) shown in the figure is only braced at the center of span.

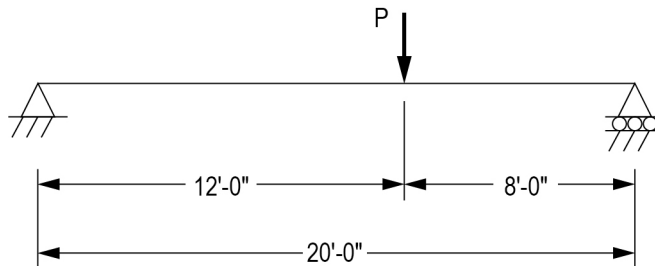
Work either the ASD or the LRFD version of the question, considering the most conservative instance.

ASD

The allowable flexural strength (ft-kips) of the beam is most nearly:

LRFD

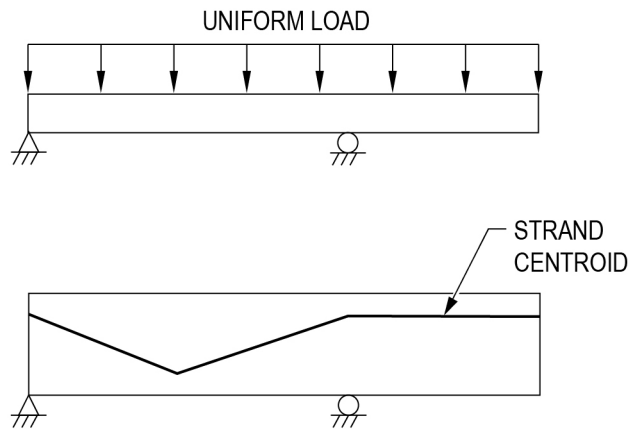
The design moment capacity ϕM_n (ft-kips) of the beam is most nearly:



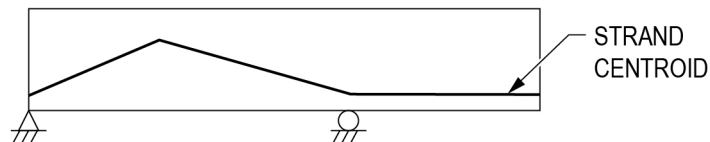
- ☐ A. ASD = 45
LRFD = 68
- ☐ B. ASD = 51
LRFD = 76
- ☐ C. ASD = 56
LRFD = 84
- ☐ D. ASD = 65
LRFD = 97

PE CIVIL: STRUCTURAL PRACTICE EXAM

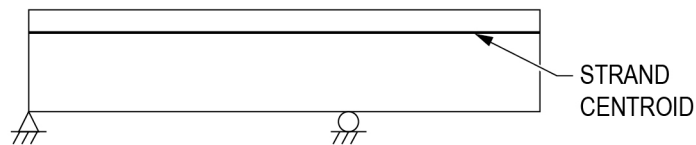
62. A loaded prestressed beam is shown in the figure. The most likely prestressing strand pattern would be:



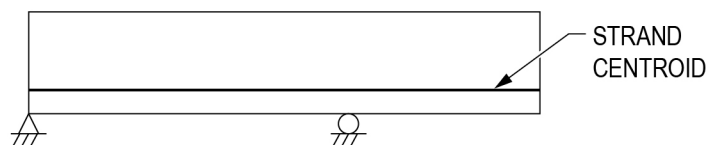
Option A



Option B



Option C



Option D

- ☐ A. Option A
- ☐ B. Option B
- ☐ C. Option C
- ☐ D. Option D

PE CIVIL: STRUCTURAL PRACTICE EXAM

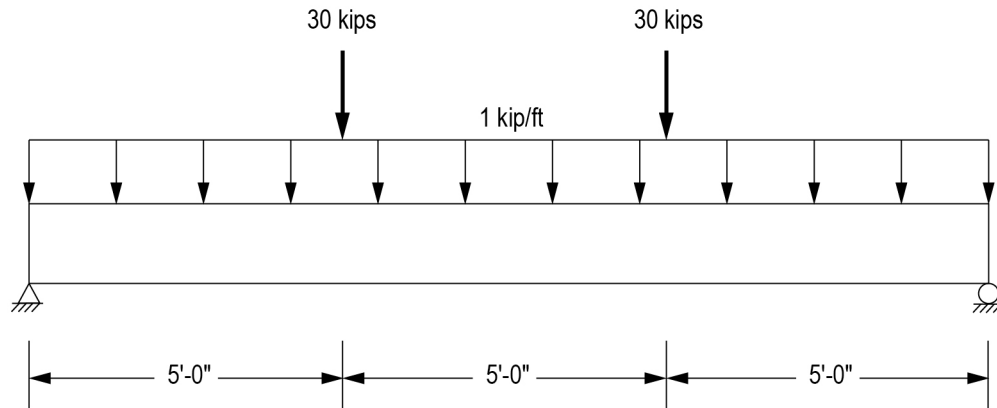
63. The following information applies to the simply supported steel beam shown in the figure:

$$E = 29,000 \text{ ksi}$$

Uniform load includes beam self-weight

Considering only deflection criteria, select all W sections from the list which meet a maximum deflection criteria of $\ell/240$.

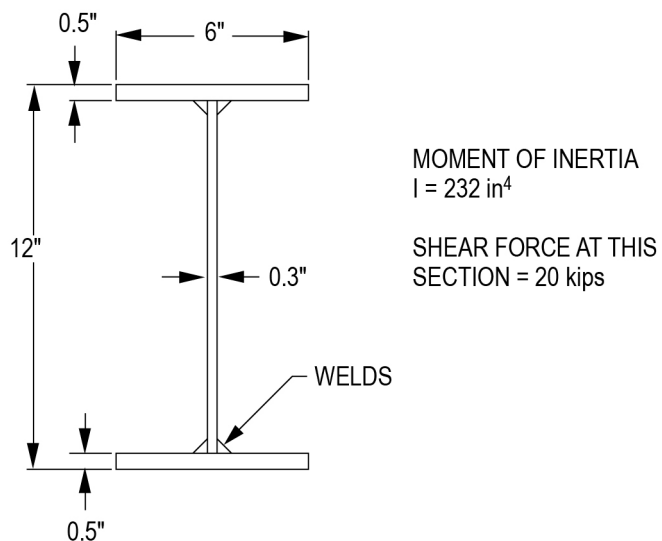
Select the **four** that apply.



- ☐ A. W12×40
- ☐ B. W12×45
- ☐ C. W12×50
- ☐ D. W14×26
- ☐ E. W14×30
- ☐ F. W14×34
- ☐ G. W16×26
- ☐ H. W16×31

PE CIVIL: STRUCTURAL PRACTICE EXAM

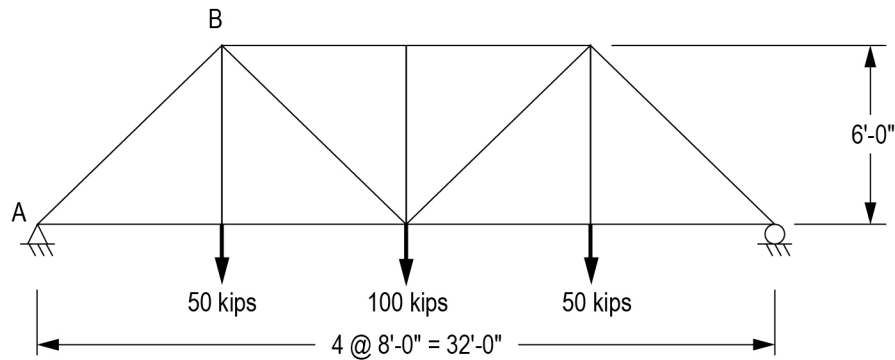
64. For the section shown, the unit shear force (kips/in.) in the welds connecting the top flange to the web (total unit shear in the two weld lines combined) is most nearly:



- ☐ A. 1.5
- ☐ B. 3.0
- ☐ C. 5.5
- ☐ D. 10.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

65. Referring to the loaded truss shown in the figure, the force (kips) in Member AB is most nearly:



- ☐ A. 100 (compression)
- ☐ B. 125 (compression)
- ☐ C. 167 (tension)
- ☐ D. 167 (compression)

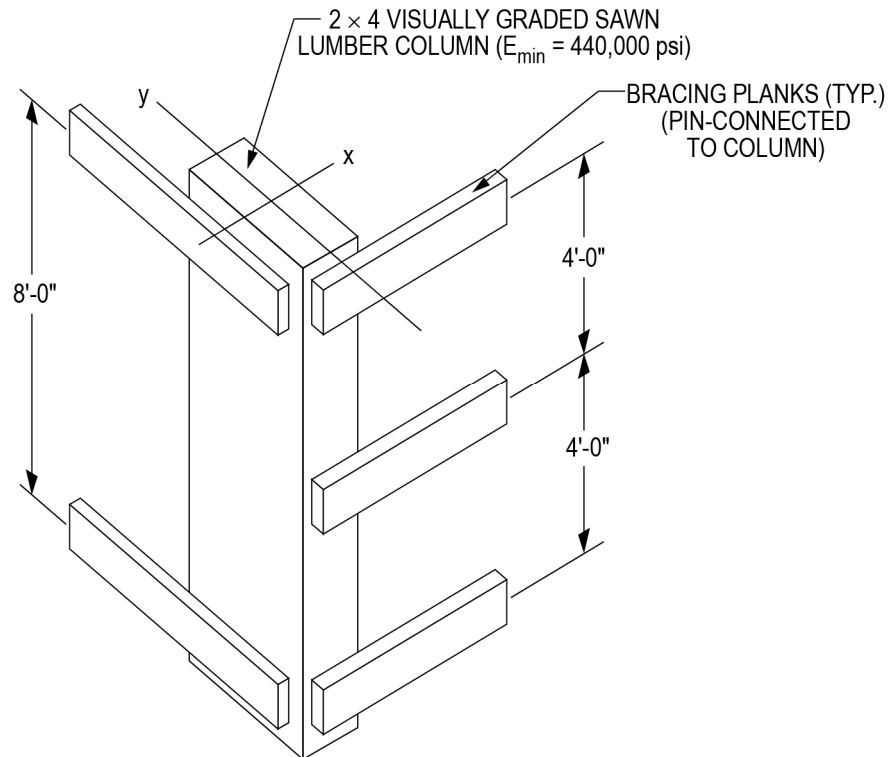
PE CIVIL: STRUCTURAL PRACTICE EXAM

66. The figure shows a timber column.

Design Code:

NDS: *National Design Specification for Wood Construction ASD/LRFD*, 2015 edition and
National Design Specification Supplement, Design Values for Wood Construction, 2015 edition.

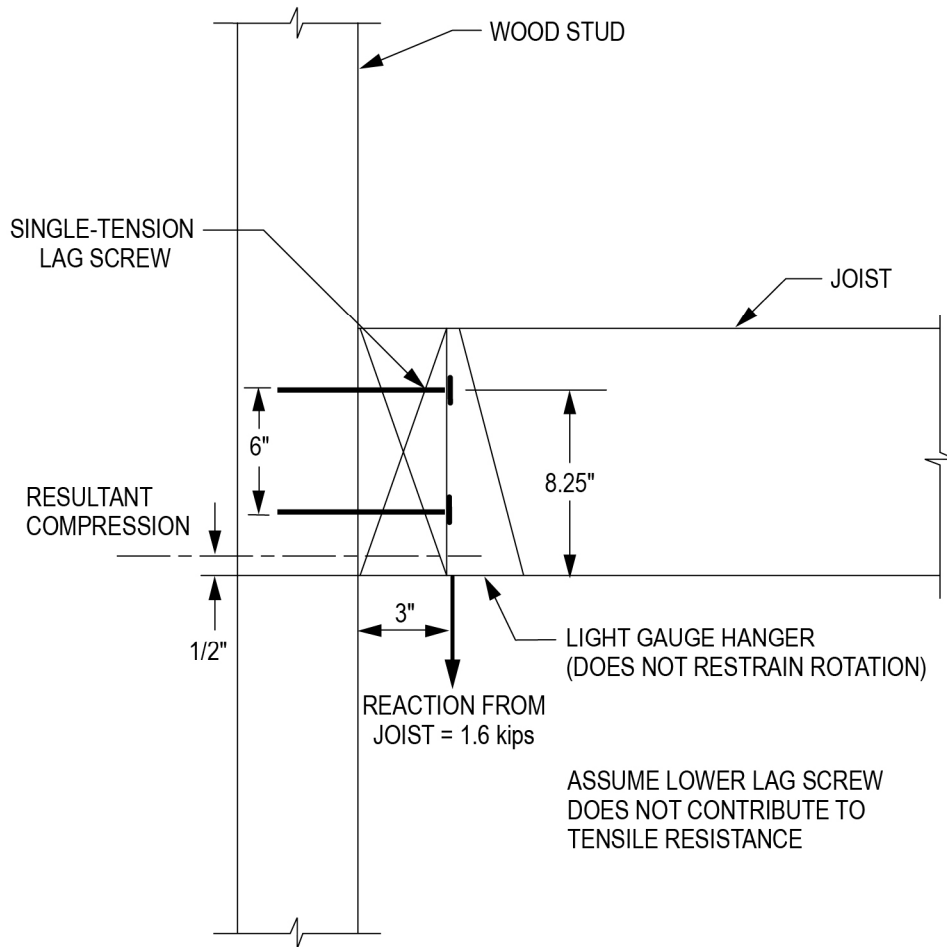
Assuming dry service, normal temperature conditions, and $C_i = 1.00$, the critical buckling design value F_{cE} (psi) using allowable stress design for the column is most nearly:



- ☐ A. 150
- ☐ B. 350
- ☐ C. 475
- ☐ D. 500

PE CIVIL: STRUCTURAL PRACTICE EXAM

67. A lag screw connection is shown in the figure. The tensile force (lb) in the top lag screw is most nearly:



- ☐ A. 582
- ☐ B. 619
- ☐ C. 800
- ☐ D. 2,400

PE CIVIL: STRUCTURAL PRACTICE EXAM

68. A connection is shown in the figure.

Design Code:

AISC: *Steel Construction Manual*, 14th edition.

Material:

A36 steel

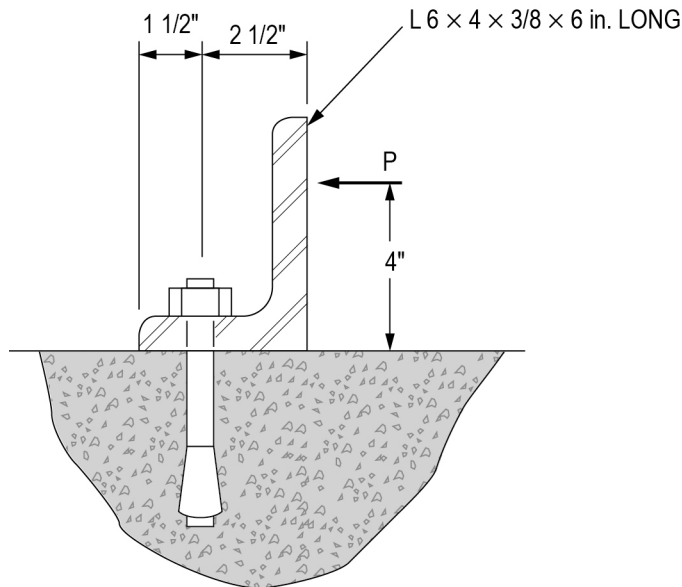
Assumptions:

The bolt is sufficient.

The concrete is sufficient.

The load P is equally distributed along the entire 6-in.-long angle.

The maximum load P (ASD) or ϕP_n (LRFD) in kips that can be carried by the angle is most nearly:



- ☐ A. ASD = 0.8
LRFD = 1.2
- ☐ B. ASD = 1.1
LRFD = 1.7
- ☐ C. ASD = 1.2
LRFD = 1.8
- ☐ D. ASD = 2.0
LRFD = 2.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

69. How is each roof diaphragm characterized? Label the correct characteristics on each diagram.

CHARACTERISTICS

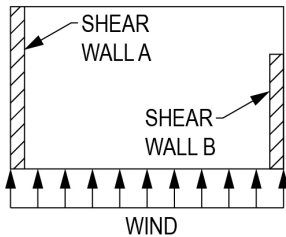
Flexible

Torsionally irregular

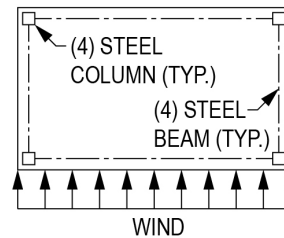
Rigid

Torsionally regular

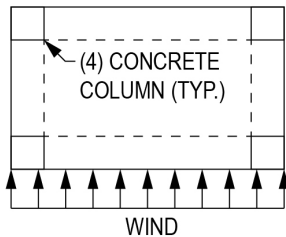
- WOOD
- DIAPHRAGM DEFLECTION CAN BE TOLERATED



- METAL DECK SUPPORTED BY STEEL BEAMS AND COLUMNS

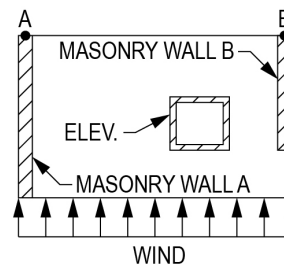


- PT CONCRETE SLAB SUPPORTED BY CONCRETE COLUMNS/BEAMS



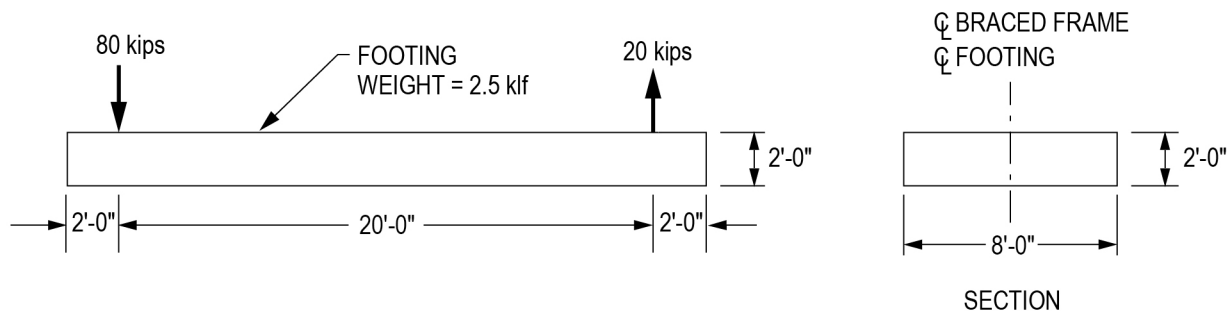
- CONCRETE SLAB SUPPORTED BY MASONRY WALLS

$$A_{\text{STORYDRIFT}} = 1.5 \left(\frac{A_{\text{STORYDRIFT}} + B_{\text{STORYDRIFT}}}{2} \right)$$



PE CIVIL: STRUCTURAL PRACTICE EXAM

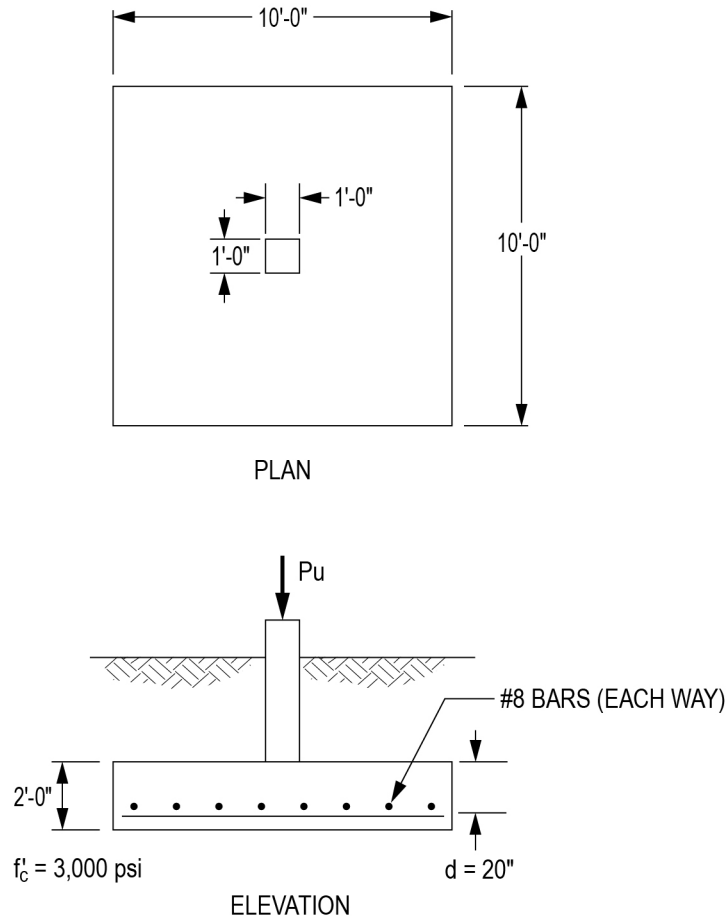
70. The concrete footing shown is subjected to loads from a braced frame. Assume the footing is rigid compared to the soil. The maximum bearing pressure (ksf) under the footing is most nearly:



- ☐ A. 0.5
- ☐ B. 1.0
- ☐ C. 2.0
- ☐ D. 2.7

PE CIVIL: STRUCTURAL PRACTICE EXAM

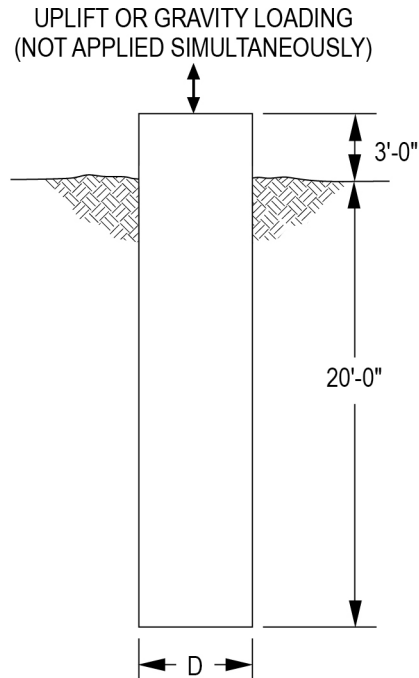
71. The figure shows a cast-in-place reinforced concrete spread footing for an interior column that is concentrically loaded. Punching shear controls the footing thickness in the design. Neglecting the shear strength of reinforcing, the design punching shear capacity of the footing (kips) per ACI 318-14, is most nearly:



- ☐ A. 870
- ☐ B. 680
- ☐ C. 560
- ☐ D. 420

PE CIVIL: STRUCTURAL PRACTICE EXAM

72. A concrete caisson is subjected to two separate loading conditions: a net uplift of 100 kips, or a net gravity load of 75 kips. The loads are not applied simultaneously. The allowable soil skin friction is 200 psf, and the allowable soil bearing pressure is 4,000 psf. Neglecting the weight of the caisson, the required caisson diameter (ft) is most nearly:



- ☐ A. 8
- ☐ B. 7
- ☐ C. 6
- ☐ D. 5

PE CIVIL: STRUCTURAL PRACTICE EXAM

73. Assuming a bearing-type connection, the load capacity (kips) for a 1-in.-diameter A307 through bolt in double shear is most nearly:

- A. ASD = 10.6
LRFD = 15.9
- B. ASD = 21.2
LRFD = 31.9
- C. ASD = 28.4
LRFD = 53.0
- D. ASD = 56.9
LRFD = 56.7

74. The following information applies to the structure shown in the figure:

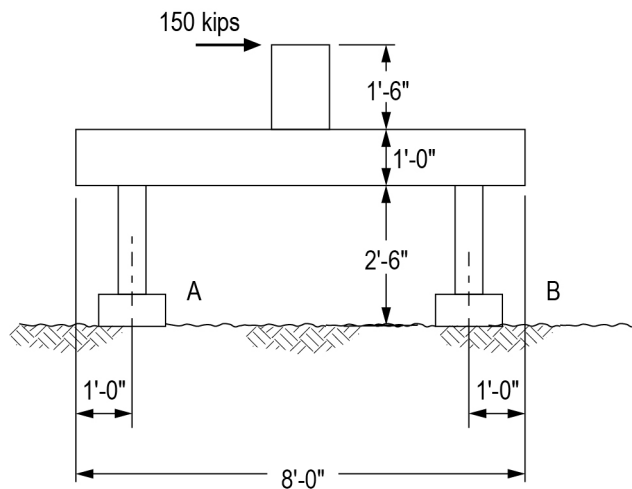
Neglect beam weight.

Beam is infinitely stiff.

Footing size is 24 in. $\times$ 24 in. at each footing.

Soil vertical modulus of subgrade, $K = 100 \text{ lb/in}^3$

The vertical settlement (in.) of the footing at B is most nearly:



- A. 2.2
- B. 1.6
- C. 1.1
- D. 0.7

PE CIVIL: STRUCTURAL PRACTICE EXAM

75. The cross section for a 50-ft-span, rectangular, prestressed beam is shown in the figure. The beam has no mild reinforcing steel.

Design Code:

PCI: *PCI Design Handbook: Precast and Prestressed Concrete*, 7th edition, 2010.

Design data for prestressing strands:

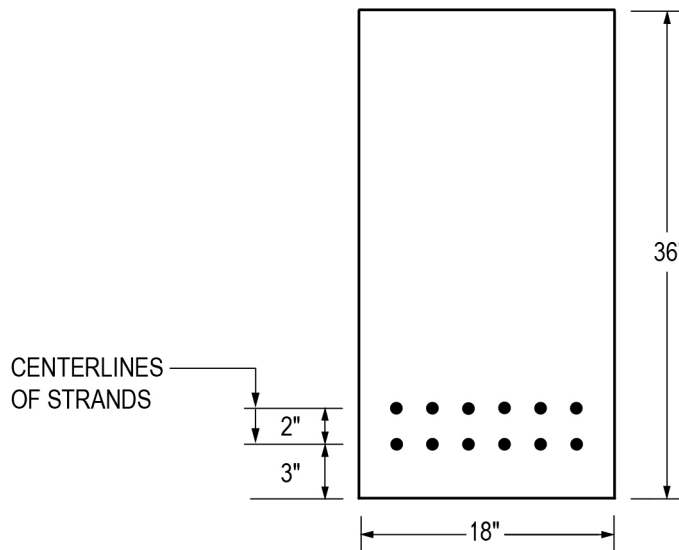
Low relaxation – 1/2-in. diameter

$f_{pu} = 270$ ksi

$A_s = 0.153$ in² per strand

Stress at release = 175 ksi per strand (after initial losses)

If the top fiber stress at the midspan of the beam due to the beam self-weight is 0.65 ksi, the total top fiber stress (ksi) at release at midspan is most nearly:



- ☐ A. 0.01 (tension)
- ☐ B. 0.50 (tension)
- ☐ C. 0.99 (compression)
- ☐ D. 2.29 (compression)

PE CIVIL: STRUCTURAL PRACTICE EXAM

76. A backfill material is described as sand-silt clay mix with plastic fines. The Unified Soil Classification is SM-SC. Per ASCE 7, the estimated design lateral soil load (psf/foot of depth) for relatively nonrigid walls is most nearly:
- ☐ A. 35
 - ☐ B. 45
 - ☐ C. 85
 - ☐ D. 100
77. A building is to be constructed on a site defined as Site Class E with a 0.2-sec mapped spectral response acceleration of 100% g. Per IBC 2015, the design spectral response acceleration at short periods is most nearly:
- ☐ A. 0.60
 - ☐ B. 0.73
 - ☐ C. 0.90
 - ☐ D. 1.60

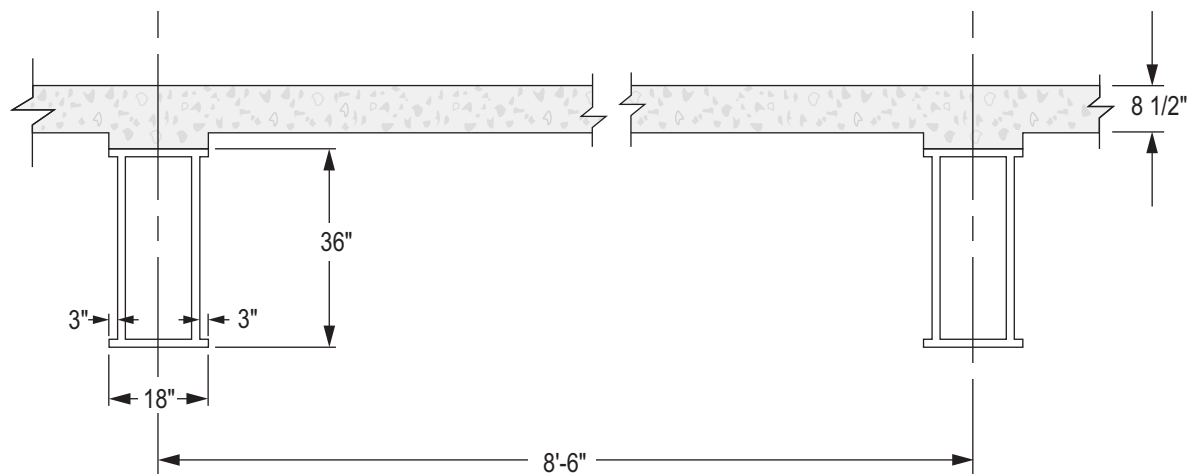
PE CIVIL: STRUCTURAL PRACTICE EXAM

78. The figure shows a cross section of a deck slab of a steel girder bridge.

Design Code:

AASHTO LRFD Bridge Design Specifications, 7th edition, 2014.

The effective span length L_{eff} to be used for the empirical design of the deck slab is most nearly:

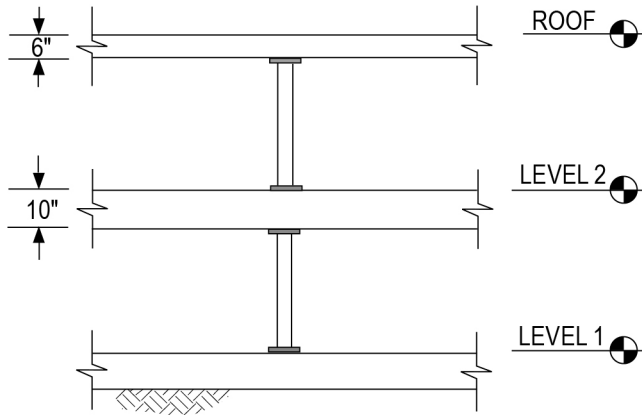


- ☐ A. 6'-9"
- ☐ B. 7'-0"
- ☐ C. 7'-3"
- ☐ D. 8'-6"

PE CIVIL: STRUCTURAL PRACTICE EXAM

79. The formwork and shoring are in place at both levels of the two-story building shown. Shores are spaced at 6 ft on center. Assume all Level 2 loads are carried by the formwork and shoring during the roof pour. During placement of concrete at the roof, the service load (lb) in a shore at Level 1 is most nearly:

NORMAL WEIGHT CONCRETE = 150 pcf
 SHORING AND FORMWORK LOAD = 12 psf
 CONSTRUCTION LIVE LOAD = 50 psf (ALL LEVELS ACTIVE)



- ☐ A. 4,930
- ☐ B. 9,860
- ☐ C. 11,650
- ☐ D. 15,420

PE CIVIL: STRUCTURAL PRACTICE EXAM

80. The first five 28-day average standard-cured concrete strength tests are shown below.

Test	28-Day Average f'_c	Date
1	4,215	March 3
2	4,160	March 7
3	3,625	March 11
4	4,010	March 12
5	4,015	March 16

The required $f'_c = 4,000$ psi. Which of the following statements is true?

- ☐ A. The concrete is unsatisfactory because the average of the last three consecutive tests is less than the required f'_c .
- ☐ B. The concrete is unsatisfactory because Test 3 did not meet the required f'_c .
- ☐ C. The concrete is satisfactory because the average of the first three tests is greater than or equal to the required f'_c .
- ☐ D. The concrete is satisfactory because Test 3 is within 500 psi of the required f'_c .

PE CIVIL: STRUCTURAL SOLUTIONS

PE CIVIL: STRUCTURAL SOLUTIONS

Detailed solutions for each question begin on the next page. Unless otherwise noted, questions can be solved using common engineering knowledge or the NCEES *PE Civil Reference Handbook*.

1	D
2	C
3	C
4	D
5	C
6	B
7	C
8	D
9	B
10	B
11	A
12	A
13	C
14	D
15	B
16	D
17	C
18	D
19	A
20	C

21	A
22	A
23	C
24	B
25	C
26	B
27	C
28	B
29	D
30	D
31	C
32	D
33	A
34	C
35	B
36	C
37	B
38	D
39	C
40	C

41	C
42	B
43	A
44	149–151
45	C
46	D
47	B
48	D
49	D
50	see solution
51	C
52	B
53	B
54	B
55	B
56	A
57	D
58	D
59	A
60	A

61	B
62	A
63	B, C, F, H
64	A
65	D
66	B
67	B
68	C
69	see solution
70	D
71	D
72	A
73	B
74	A
75	A
76	C
77	A
78	C
79	C
80	A

PE CIVIL: STRUCTURAL SOLUTIONS

1. Horizontal length of side slope = $14 \times \frac{3}{2} = 21.0 \text{ ft}$
 Slope length = $\sqrt{(14)^2 + (21)^2} = 25.24 \text{ ft}$
 Cross-sectional area of lining = $\left[(2 \times 25.24) + 9\right] \frac{7}{12} = 34.70 \text{ ft}^2$
 Volume of lining = $\frac{(34.70 \times 227)}{27} = 291.7 \text{ yd}^3$
 Delivered volume = $291.7 \text{ yd}^3 \times 1.12_{(\text{waste})} = 327 \text{ yd}^3$

THE CORRECT ANSWER IS: D

2. $D = \frac{\$75,000 - \$10,000}{10}$
 $D = \$6,500$
 Book value after 8 years = $\$75,000 - (8)(\$6,500) = \$23,000$

THE CORRECT ANSWER IS: C

3. Crew cost = $2(\$50/\text{hr}) = \$100/\text{hr}$
 Days allowed = $\frac{\$4,000}{(8 \text{ hr/day})(\$100/\text{hr})} = 5 \text{ days}$

THE CORRECT ANSWER IS: C

4. Activities: $(7) + (4) + (5)$
 Days: $30 + 10 + 10 = 50 \text{ days}$

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

$$5. \quad 1,000 \text{ kN} = 1,000 \text{ kN} \times \frac{1 \text{ ton}}{8.896444 \text{ kN}} = 112.4 \text{ tons}$$

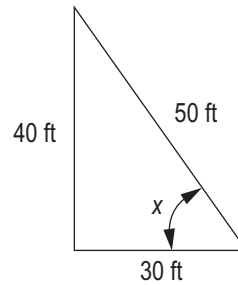
$$150 \text{ tons} > 112.4 \text{ tons}$$

THE CORRECT ANSWER IS: C

$$6. \quad \tan(x) = \frac{40}{30} \quad x = 53.13^\circ$$

$$\cos(53.13^\circ) \times 100 \text{ ft} = 60 \text{ ft}$$

$$60 \text{ ft} - 35 \text{ ft} = 25 \text{ ft}$$

**THE CORRECT ANSWER IS: B**

$$7. \quad w = (20 \text{ lb/ft}^2)(8 \text{ ft}) = 160 \text{ lb/vertical ft per brace location}$$

$$\sum M_a = 0$$

$$\sum M_a = (160 \text{ lb/ft})(16 \text{ ft})(16 \text{ ft}/2) - 10 \text{ ft} (R_x) = 0$$

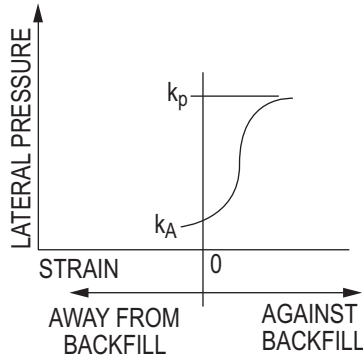
$$R_x = 2,048 \text{ lb}$$

$$\text{Axial load in brace} = \frac{(2,048)\sqrt{2}}{1} = 2,896 \text{ lb}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

8. The wall translation (or strain) required to achieve the passive state is at least twice that required to reach the active state.



THE CORRECT ANSWER IS: D

9. The solution is based on the knowledge that consolidation settlement is the result of the expulsion of pore water from saturated soil due to imposed load. Therefore, the volume of the wick drain effluent (water) to be treated equals the consolidation settlement volume over the affected site area, and is computed as follows:

$$\text{Affected area} = 21.5 \text{ acres} \times 43,560 \text{ ft}^2/\text{acre} = 936,540 \text{ ft}^2$$

$$\text{Mean consolidation settlement over affected area} = 22 \text{ in.} = 1.83 \text{ ft}$$

$$\text{Settlement volume} = \text{effluent volume} = 936,540 \text{ ft}^2 \times 1.83 = 1,713,868 \text{ ft}^3$$

$$\text{Convert to gal: } 1,713,868 \text{ ft}^3 \times 7.48 \text{ gal/ft}^3 = 12,819,733 \text{ gal}$$

$$\text{Cost for effluent treatment and disposal} = 12,819,733 \text{ gal} \times \$0.25/\text{gal} = \$3,204,934$$

THE CORRECT ANSWER IS: B

10. Effective vertical stress at Point A, σ'_v
- $$= 10 \text{ ft} \times 120 \text{ pcf} + 5 \text{ ft}(120 \text{ pcf} - 62.4 \text{ pcf}) + 7 \text{ ft}(110 \text{ pcf} - 62.4 \text{ pcf})$$
- $$= 1,200 \text{ psf} + 288 \text{ psf} + 333 \text{ psf}$$
- $$= 1,821 \text{ psf}$$

THE CORRECT ANSWER IS: B

11. The ultimate bearing capacity would be based on buoyant unit weight, also referred to as the effective unit weight.

$$\text{Effective unit weight} = \text{saturated unit weight} - \text{unit weight of water}$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

12. The long-term settlement for Case I is less than Case II because clay is subject to long-term settlement.

THE CORRECT ANSWER IS: A

13. The minimum factor of safety for permanent slopes is 1.5. Other references use a factor of safety greater than or equal to 1.3, but of the options presented, 1.5 is the closest.

THE CORRECT ANSWER IS: C

14. Because the structure is cantilevered, in addition to the wind, dead load and live load will contribute to uplift.

THE CORRECT ANSWER IS: D

15. By inspection, Member b = 0 kips, and Member c = 100 kips.

THE CORRECT ANSWER IS: B

16. Beam stress, $f = M/S$, where $M = wL^2/8$ and $S = bh^2/6$.
S is equal for both beams, but M varies because it depends on beam length.

$$\text{Beam 1 (shorter beam): } M_1 = wL^2/8$$

$$\text{Beam 2 (longer beam): } M_2 = w(2L)^2/8 = 4wL^2/8$$

M_2 is four times greater than M_1 . Therefore, the maximum bending stress is four times greater in the longer beam.

THE CORRECT ANSWER IS: D

17. Uniform load: $V = \frac{wL}{2} = \frac{1(30)}{2} = \frac{30 \text{ kips}}{2} = 15 \text{ kips}$

$$\text{Point load: } V = \frac{P}{2} = 15 \text{ kips}$$

$$P = 2(15) = 30 \text{ kips}$$

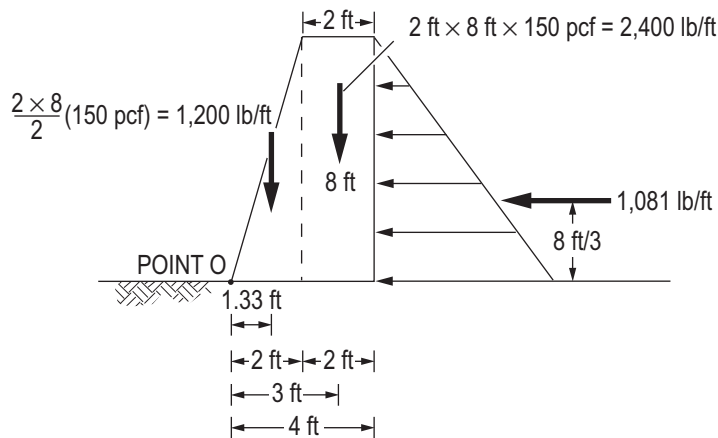
THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

18. I_x is maximum for this section by inspection, or calculate $I_x \approx \sum Ad^2$ for each section.

THE CORRECT ANSWER IS: D

19. $\phi = 32^\circ$ $K_a = \tan^2(45 - \phi/2) = 0.307$
 $\gamma_t = 110 \text{ pcf}$ $P_a = (0.5)(110)(8)^2(0.307) = 1,081 \text{ lb/ft}$
 $M_a = (1,081)(8/3) = 2,883 \text{ ft-lb/ft}$
 $(2)(8)(150)(1)(3) = 7,200 \text{ ft-lb/ft}$
 $(1/2)(2)(8)(150)(1)(2)(2/3) = 1,600 \text{ ft-lb/ft}$ } total = 8,800 ft-lb/ft
 $SF = 8,800/2,883 = 3.05$



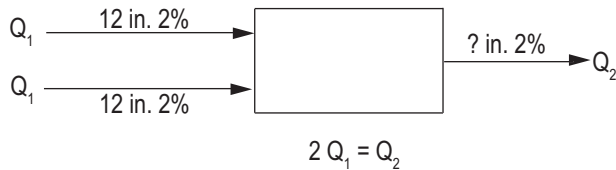
THE CORRECT ANSWER IS: A

20. $Q = VA = \left\{ \frac{1.49}{n} R^{2/3} S^{1/2} \right\} A$
 $= \left\{ \frac{1.49}{0.022} \left[\frac{(1.5 \text{ ft} \times 4 \text{ ft})}{4 \text{ ft} + 2(1.5 \text{ ft})} \right]^{2/3} (0.002)^{1/2} \right\} (1.5 \text{ ft} \times 4 \text{ ft})$
 $= 16.4 \text{ cfs}$

$$\text{Volume} = 25 \text{ acre-ft} \times \frac{43,560 \text{ ft}^3}{1 \text{ acre-ft}} = 1.089 \times 10^6 \text{ ft}^3$$

$$\text{Time} = \frac{1.089 \times 10^6 \text{ ft}^3}{16.4 \text{ ft}^3/\text{sec}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 18.5 \text{ hours}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS**21.**

$$2[V_1 A_1] = [V_2 A_2]$$

$$2 \left[\left(\frac{1.49}{n} \right) (A_1) R_1^{2/3} S^{1/2} \right] = \left[\left(\frac{1.49}{n} \right) (A_2) R_2^{2/3} S^{1/2} \right]$$

$$2 \left[(A_1) \left(\frac{A_1}{P_1} \right)^{2/3} \right] = \left[(A_2) \left(\frac{A_2}{P_2} \right)^{2/3} \right]$$

$$A_1 = \frac{\pi D^2}{4} = \frac{\pi (1)^2}{4} = 0.785 \text{ ft}^2$$

$$P_1 = \pi(D) = \pi(1) = 3.14 \text{ ft}$$

$$2 \left[(0.785) \left(\frac{0.785}{3.14} \right)^{2/3} \right] = \left[\left(\frac{\pi D_2^2}{4} \right) \left(\frac{\pi (D_2)^2}{4 (\pi D_2)} \right)^{2/3} \right]$$

$$0.623 = \left(\frac{\pi D_2^2}{4} \right) \left(\frac{D_2}{4} \right)^{2/3}$$

$$= \pi \left(\frac{D_2^2}{4} \right) \left(\frac{D_2}{4} \right)^{2/3}$$

$$= \pi (D_2)^{8/3} \left(\frac{1}{4} \right) \left(\frac{1}{4} \right)^{2/3}$$

$$0.623 = 0.311 (D_2)^{8/3}$$

$$\left(\frac{0.623}{0.311} \right)^{3/8} = D_2$$

$$D_2 = 1.297 \text{ ft} \times \frac{12 \text{ in.}}{\text{ft}} = 15.6 \text{ in.} \approx 16 \text{ in.}$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

22. According to the arithmetic mean method, the average precipitation is simply the average of all the rainfall gauges.

$$\begin{aligned}\text{Average precipitation} &= (2.1 + 3.6 + 1.3 + 1.5 + 2.6 + 6.1 + 5.1 + 4.8 + 4.1 + 2.8 + 3.0)/11 \\ &= 3.4 \text{ in.}\end{aligned}$$

THE CORRECT ANSWER IS: A

23. From the IDF curve, read a rainfall intensity of 3.5 in./hr for a 50-year frequency rainfall with a 60-min duration.

From the table, the runoff coefficient for a downtown area is 0.70–0.95. For the maximum runoff rate, use the high value of 0.95.

$$Q = CiA = 0.95 \times 3.5 \text{ in./hr} \times 90 \text{ ac}$$

$$Q = 300 \text{ cfs}$$

THE CORRECT ANSWER IS: C

24. $\text{Time} = \frac{V}{Q}$

$$V = 400,000 \text{ gal} \times \frac{\text{ft}^3}{7.48 \text{ gal}} = 53,476 \text{ ft}^3$$

$$Q = 1.5 \text{ ft}^3/\text{sec}$$

$$\text{Time} = \frac{53,476 \text{ ft}^3}{1.5 \text{ ft}^3/\text{sec}} \times \frac{1 \text{ hr}}{3,600 \text{ sec}} = 9.9 \text{ hours}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

25. The Darcy–Weisbach equation is $h_f = f \frac{L}{D} \frac{V^2}{2g}$

where

h_f = headloss, ft

f = friction factor, unitless

L = length, ft

D = diameter of pipe, ft

V = velocity, ft/sec

g = gravitational constant, 32.2 ft/sec²

Substituting gives

$$5 \text{ ft} = 0.0115 \times \frac{1,650 \text{ ft}}{3.0 \text{ ft}} \times \frac{V^2}{2 \times 32.2 \text{ ft/sec}^2}$$

$$V^2 = 50.91 \text{ ft}^2/\text{sec}^2$$

$$V = 7.135 \text{ ft/sec}$$

$$Q = VA = V \times \frac{\pi}{4} D^2 = 7.135 \text{ ft/sec} \times \frac{\pi}{4} (3.0 \text{ ft})^2$$

$$Q = 50 \text{ cfs}$$

THE CORRECT ANSWER IS: C

26. $z_1 + \frac{P_1}{\gamma} + \frac{v_1^2}{2g} = z_2 + \frac{P_2}{\gamma} + \frac{v_2^2}{2g}$

$$z_1 = z_2$$

Since $A_1 > A_2$, $v_1 < v_2$.

$$\therefore \frac{v_1^2}{2g} < \frac{v_2^2}{2g}$$

so $P_1 > P_2$ to balance

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

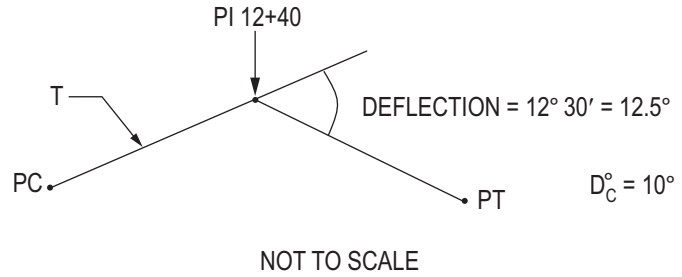
27. $R = 5,729.648/D_C^\circ$
 $= 5,729.648/10 = 572.96 \text{ ft}$
 $T = R \tan\left(\frac{1}{2}\Delta\right) = R \tan(6.25^\circ)$
 $= 572.96 (\tan 6.25^\circ)$
 $= 572.96 (0.1095178)$
 $= 62.75 \text{ ft}$

Station PC = Station PI - T
 $= (12 + 40) - 62.75$
 $= 11 + 77.25$

Station PT = Station PC + length of curve

Length of curve = $L = 100 \Delta/D_C^\circ$
 $= 100(12.5)/10 = 125 \text{ ft}$

Station PT = Station PC + 125 ft = $(11 + 77.25) + 125 = 13 + 02.25$



THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

28. $L = KA$
 $K = L/A$

L = length of vertical curve, ft

A = algebraic difference in grades, percent ($g_2 - g_1$)

Given: VPC = 12+00

VPI = 13+50

VPT = 15+00

$g_1 = -2.30\%$

$g_2 = +3.00\%$

$L = 300$ ft

$$K = \frac{L}{A} = \frac{300}{3 - (-2.3)} = 56.60 \text{ ft/percent for the vertical curve.}$$

The length from Station 14+00 to Station 15+00 = 100 ft

$$K = \frac{L}{A}$$

$$A = \frac{L}{K} = \frac{100}{56.60} = 1.77\%$$

$$A = g_2 - g_1$$

Tangent slope at Station 14+00 = g_1

$$g_1 = g_2 - A = 3.00\% - 1.77\% = 1.23\%$$

Alternate solution:

Y = elevation at a point X ft from VPC

Y' = slope at a point X ft from VPC

$$X = [14 + 00] - [12 + 00] = 200 \text{ ft}$$

g_1 = slope 1 in ft/ft

g_2 = slope 2 in ft/ft

L = length of vertical curve, ft

$$Y = Y_{VPC} + g_1 X + \left(\frac{g_2 - g_1}{2L} \right) X^2$$

$$Y' = g_1 + \left(\frac{g_2 - g_1}{L} \right) X$$

$$Y' = -0.023 + \left(\frac{0.03 - (-0.023)}{300} \right) 200 = 0.0123 \text{ ft/ft or } 1.23\%$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

29.
$$\text{AADT} = \frac{\Sigma (\text{Jan. through Dec.})}{12}$$
$$= 833,200 / 12 = 69,433$$
$$\Sigma (\text{June through Aug.}) = 77,300$$
$$\begin{array}{r} 78,950 \\ 77,200 \\ \hline 233,450 / 3 = 77,817 \end{array}$$

Seasonal factor for June through August

$$= 77,817 / 69,433$$
$$= 1.121$$

THE CORRECT ANSWER IS: D

30. The commonly used soil classification systems for engineering applications are USCS and AASHTO. Both of these systems use gradation and Atterberg limits as two of the criteria.

THE CORRECT ANSWER IS: D

31. The Standard Penetration Test (SPT) N-value provides an indication of the relative density of cohesionless soils.

THE CORRECT ANSWER IS: C

32. An early-strength concrete is needed with a minimum compressive strength of 3,500 psi. To achieve the requirements, a Type III cement and chemical accelerators would be necessary.

THE CORRECT ANSWER IS: D

33. **THE CORRECT ANSWER IS: A**

PE CIVIL: STRUCTURAL SOLUTIONS

34. $\text{Area} = \pi d^2/4 = 28.3 \text{ in}^2$

Compressive stress = axial load/area

Sample 1 $f'_c = \frac{65,447}{28.3} = 2,313 \text{ psi}$

Sample 2 $f'_c = \frac{63,617}{28.3} = 2,248 \text{ psi}$

Sample 3 $f'_c = \frac{69,872}{28.3} = 2,469 \text{ psi}$

Average $= \frac{(2,313 + 2,248 + 2,469)}{3} = 2,343 \text{ psi}$

THE CORRECT ANSWER IS: C

35. $\text{Total density } (\gamma) = \frac{W}{V} = \frac{W_s + W_w}{V_s + V_w + V_a}$

where γ = total density

W = total weight

V = total volume

W_s = weight soil

W_w = weight of water

V_s = volume of soil

V_w = volume of water

V_a = volume of air

$\gamma = \frac{9.11 \text{ lb} - 4.41 \text{ lb}}{0.03 \text{ ft}^3} = 156.67 \text{ lb/ft}^3 \text{ (pcf)}$

Dry unit weight of soil $(\gamma_d) = \frac{\gamma}{1 + w}$

where w = moisture content

$\gamma_d = \frac{156.67 \text{ pcf}}{1 + 0.115} = 140.51 \text{ pcf}$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

36. Use Average End Area Method.

Stationing	Excavation (yd ³)	Embankment (yd ³)
1+00 to 2+00	$\frac{50+150}{2} \times \frac{100}{27} = 370$	
2+00 to 3+00	$\frac{50+0}{2} \times \frac{100}{27} = 93$	$\frac{0+40}{2} \times \frac{100}{27} = 74$
Total	463	74

Net excess excavated material = $463 - 74 = 389 \text{ yd}^3$

THE CORRECT ANSWER IS: C

37. Existing:

$$\Delta H = (2 + 88.4) - (0 + 23.0) = 288.4 - 23.0 = 265.4 \text{ ft}$$

$$\Delta V = 630.32 - 609.39 = 20.93 \text{ ft}$$

New:

$$\Delta H = (1 + 15.0) - (0 + 23.0) = 115.0 - 23.0 = 92 \text{ ft}$$

$$\Delta V = \frac{92}{265.4} \times 20.93 = 7.26 \text{ ft}$$

$$\text{Inv Elev.} = 630.32 - 7.26 = 623.06 \text{ ft}$$

The top of the pipe will be above the invert elevation by $(60 \text{ in.} - 6 \text{ in.})/12 \text{ in./ft} = 4.50 \text{ ft}$

$$623.06 + 4.50 = 627.56 \text{ ft}$$

THE CORRECT ANSWER IS: B

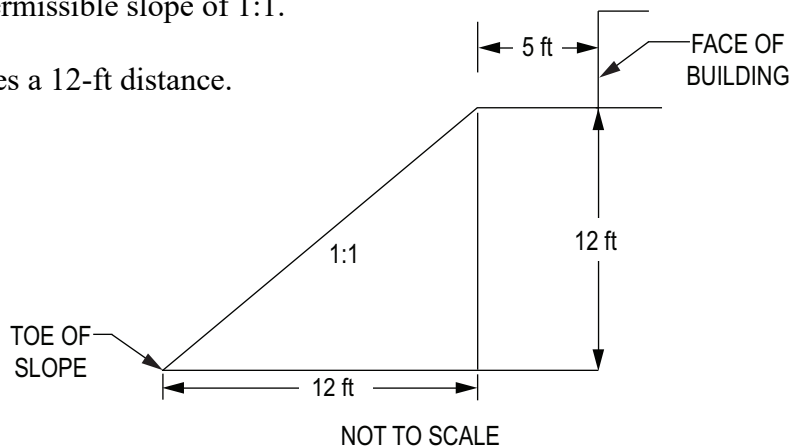
38. Rushing erosion is not a stormwater erosion classification.

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

39. Type B soil has a maximum permissible slope of 1:1.

Therefore, a 12-ft depth requires a 12-ft distance.

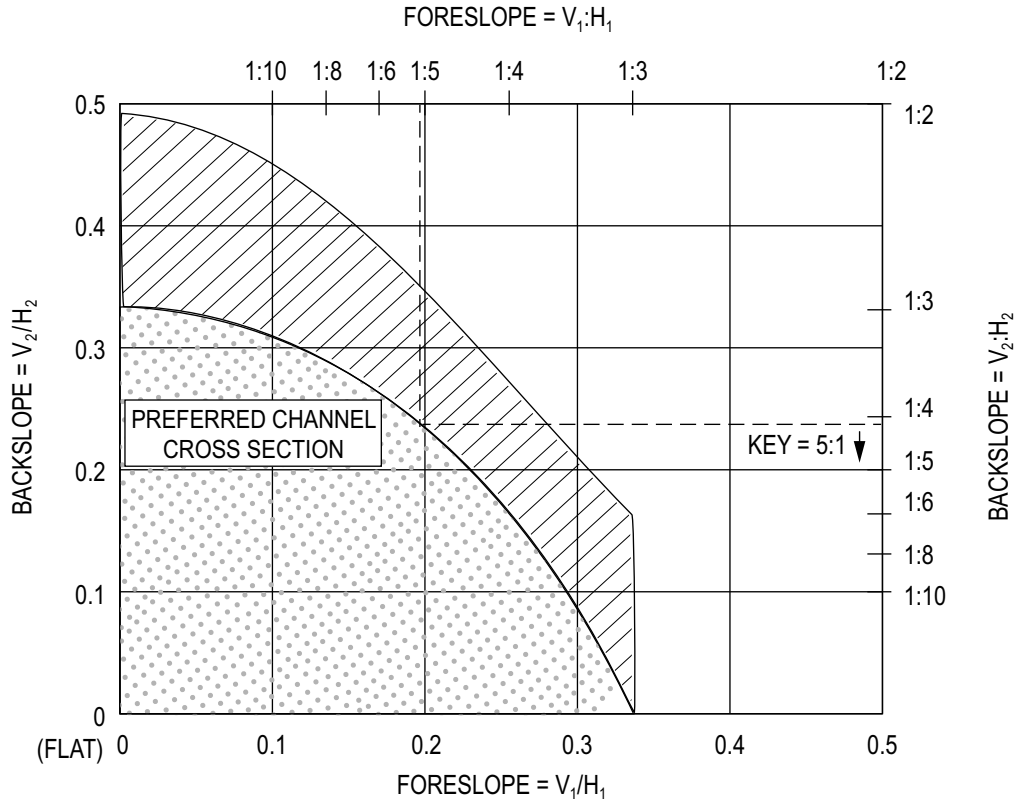


Because there is a 5-ft perimeter strip, the minimum distance from the toe of the slope to the face of the structure = 12 ft + 5 ft = 17 ft.

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

40.



This area is applicable to all Vee ditches, rounded channels with a bottom width less than 2.4 m [8 ft], and trapezoidal channels with bottom widths less than 1.2 m [4 ft].

This area is applicable to rounded channels with bottom width of 2.4 m [8 ft] or more and to trapezoidal channels with bottom widths equal to or greater than 1.2 m [4 ft].

Adapted from AASHTO *Roadside Design Guide*, 4th edition, 2011.

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

41. $p_f = 0.7 C_e C_t I_s p_g$ (Eq. 7.3-1)

where $C_e = 0.9$ (Table 7-2)

$C_t = 1.2$ (Table 7-3)

$I_s = 0.8$ (Table 1.5-2)

$p_g = 20$ psf

$$p_f = (0.7)(0.9)(1.2)(0.8)(20)$$

$$= 12.1 \text{ psf}$$

$$p_s = C_s p_f \text{ (Eq. 7.4-1) and } C_s = 1.0 \text{ (Fig. 7-2)}$$

$$= 1.0(12.1) = 12.1$$

Unbalanced snow load $= I_s p_g$ (Par. 7.6.1)

$$= 0.8(20)$$

$$= 16 \text{ psf}$$

16 psf controls

THE CORRECT ANSWER IS: C

42. $\frac{60(200)}{2} = 6 \text{ kips}$

THE CORRECT ANSWER IS: B

43. Reference: IBC 2015/ASCE 7-10. IBC 2015 refers to ASCE 7-10.
Per ASCE 7-10, 11.7, (Seismic Design Category A)

$$F = 0.01w \quad w = \text{dead load} \quad (\text{Eq. 1.4-1})$$

$$= 0.01(200,000)$$

$$= 2,000 \text{ lb}$$

THE CORRECT ANSWER IS: A

44. Maximum moment occurs when load is at beam midspan.

$$M = \frac{PL}{4} = \frac{(20 \text{ kips})(30 \text{ ft})}{4} = 150 \text{ ft-kips}$$

THE ACCEPTABLE RANGE IS: 149–151

PE CIVIL: STRUCTURAL SOLUTIONS

45. Take moments about Support B:

$$R_a \times 24 - 6.4 \times 24 - 2 \times 24^2/2 + 20 + 1.55 \times 8^2/2 + 1 \times 8 = 0$$

$$R_a = 28 \text{ kips}$$

THE CORRECT ANSWER IS: C

46. Calculate reactions by taking moments about Support A:

$$(6 \text{ kips})(5 \text{ ft}) + (1 \text{ kip/ft})(20 \text{ ft})(20 \text{ ft}/2) = R_B(20 \text{ ft})$$

$$R_B = 11.5 \text{ kips}; R_A = 14.5 \text{ kips}$$

Location of maximum moment is where shear diagram = 0

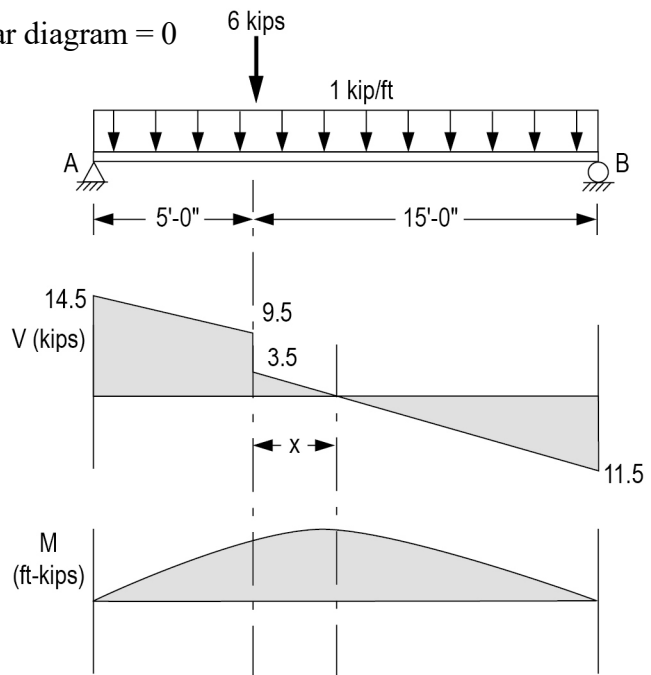
$$\frac{3.5 \text{ kips}}{x} = \frac{11.5 \text{ kips}}{(15 \text{ ft} - x)}$$

$$3.5(15 - x) = 11.5x$$

$$x = 3.5 \text{ ft}$$

Distance from right end:

$$20 \text{ ft} - (5 \text{ ft} + 3.5 \text{ ft}) = 11.5 \text{ ft}$$



THE CORRECT ANSWER IS: D

47. $FEM_{BA} = \frac{Pa^2b}{L^2} = \frac{(2)(12)^2(8)}{20^2} = 5.76 \text{ ft-kips}$

$$FEM_{BC} = \frac{wL^2}{12} = \frac{(1)(15)^2}{12} = 18.75 \text{ ft-kips}$$

$$18.75 - 5.76 = 12.99 \text{ ft-kips}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

48. $n = \frac{E_{\text{steel}}}{E_{\text{wood}}} = \frac{29 \times 10^6}{1.6 \times 10^6} = 18.125$

$A'_{\text{steel}} = nwh = (18.125)(6)(0.125) = 13.6 \text{ in}^2$

$A_{\text{wood}} = wh = (6)(8) = 48 \text{ in}^2$

$A = A'_{\text{steel}} + A_{\text{wood}} = 13.6 + 48 = 61.6 \text{ in}^2$

$\Sigma \bar{y}A = (\bar{y}_{\text{steel}} \times A'_{\text{steel}}) + (\bar{y}_{\text{wood}} \times A_{\text{wood}}) = (0.0625)(13.6) + (4.125)(48) = 198.85 \text{ in}^3$

$c = \bar{y} = \frac{\Sigma \bar{y}A}{A} = \frac{198.85}{61.6} = 3.23 \text{ in.}$

$I_{\text{wood}} = \frac{b \times d^3}{12} + [A_{\text{wood}} \times (\bar{y}_{\text{wood}} - \bar{y})^2] = \frac{(6)(8)^3}{12} + (48)(4.125 - 3.23)^2 = 294.4 \text{ in}^4$

$I_{\text{steel}} = \frac{b' \times d^3}{12} + [A'_{\text{steel}} \times (\bar{y} - \bar{y}_{\text{steel}})^2]$

$= \frac{(18.125)(6)(0.125)^3}{12} + (13.6)(3.23 - 0.0625)^2 = 136.5 \text{ in}^4$

$I = I_{\text{wood}} + I_{\text{steel}} = 294.4 + 136.5 = 430.9 \text{ in}^4$

$S_{\text{bottom}} = \frac{I}{c} = \frac{430.9}{3.23} = 133 \text{ in}^3$

$M = \frac{P \times a \times b}{L} = \frac{(6)(4)(6)}{10} = 14.4 \text{ ft-kips}$

$f'_b = \frac{M}{S} = \frac{(14.4)(12 \text{ in./ft})}{133} = 1.3 \text{ ksi}$

$f_b = f'_b \times n = (1.3)(18.125) = 23.5 \text{ ksi}$

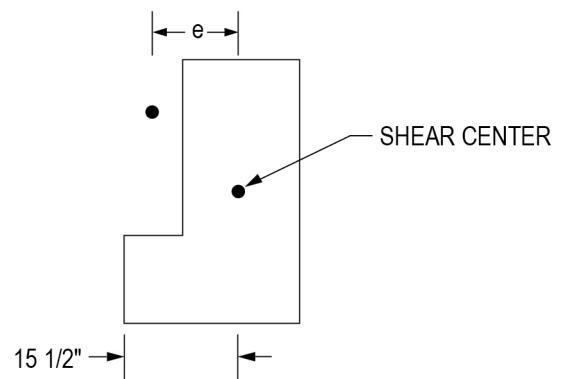
THE CORRECT ANSWER IS: D

49. $15.5 \text{ in.} - 6 \text{ in.} = 9.5 \text{ in.}$

$W_{u/DT} = [1.2(10 \text{ psf}) + 1.6(50 \text{ psf})]15 \text{ ft} = 1,380 \text{ plf}$

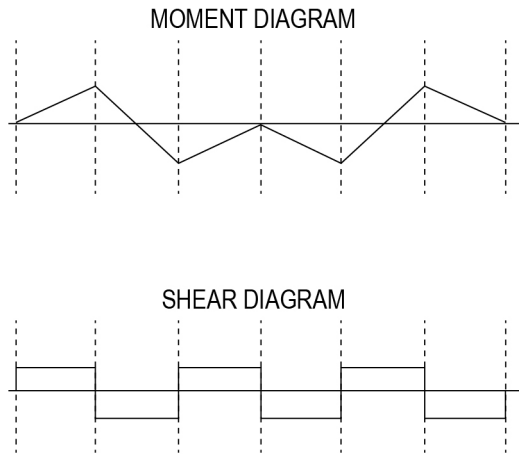
$R_{u/DT} = \frac{1}{2}(1.38 \text{ klf})(60 \text{ ft}) = 41.4 \text{ kips}$

$T_u = 41.4 \text{ kips} \left(\frac{9.5 \text{ in.}}{12 \text{ in./ft}} \right) = 32.8 \text{ ft-kips}$

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

50. Referring to the beam shown in the question, there is a zero moment at the hinge.



THE CORRECT DIAGRAMS ARE SHOWN ABOVE.

51.
$$M_{\max} = \frac{(200 \text{ lb/ft})(20 \text{ ft})^2}{8} + \frac{(1,000 \text{ lb})(20 \text{ ft})}{4}$$

$$= 10,000 \text{ ft-lb} + 5,000 \text{ ft-lb} = 15,000 \text{ ft-lb}$$

$$I = \frac{(4 \text{ in.})(12 \text{ in.})^3}{12} = 576 \text{ in}^4$$

$$f_b = \frac{Mc}{I}$$

$$= \frac{(15,000 \text{ ft-lb})(6 \text{ in.})}{576 \text{ in}^4} (12 \text{ in./ft})$$

$$f_b = 1,875 \text{ psi}$$

THE CORRECT ANSWER IS: C

52.
$$I = \frac{bd^3}{12} - \frac{b_1d_1^3}{12} = \frac{8(12)^3 - 4(8)^3}{12} = 981 \text{ in}^4$$

$$V = 1,000 \text{ plf} \left(\frac{10 \text{ ft}}{2} \right) + \left(\frac{4,000 \text{ lb}}{2} \right) = 7,000 \text{ lb}$$

$$V_{\text{horizontal}} = \frac{VQ}{IB} = \frac{7,000 \text{ lb} (85 \text{ in}^3)}{981 \text{ in}^4 (2 \times 2 \text{ in.})} = 151 \text{ psi}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

53. $M_S = (20 \text{ kips})(12 \text{ in.} - 5 \text{ in.}) = 140 \text{ in.-kips}$

$N_S = 20 \text{ kips}$

$$\sigma_{\text{flexure}} = \frac{(140 \text{ in.-kips})(5 \text{ in.})}{90 \text{ in}^4} = 7.8 \text{ ksi} \begin{cases} \text{Top tension} \\ \text{Bottom compression} \end{cases}$$

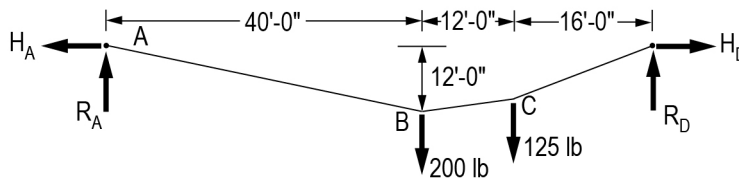
$$\sigma_{\text{axial}} = \frac{20 \text{ kips}}{7 \text{ in}^2} = 2.9 \text{ ksi tension}$$

Max compressive stress (at bottom):

$$-7.8 \text{ ksi} + 2.9 \text{ ksi} = -4.9 \text{ ksi}$$

THE CORRECT ANSWER IS: B

54.



$$R_A = \frac{200 \text{ lb} \times 28 \text{ ft} + 125 \text{ lb} \times 16 \text{ ft}}{68 \text{ ft}} = 111.76 \text{ lb}$$

$$R_D = \frac{200 \text{ lb} \times 40 \text{ ft} + 125 \text{ lb} \times 52 \text{ ft}}{68 \text{ ft}} = 213.24 \text{ lb}$$

$$H_A = \frac{40 \text{ ft}}{1 \text{ ft}} \times R_A = 4,470.4 \text{ lb}$$

$$H_A = H_D$$

$$F_{CD} = \sqrt{213.24^2 + 4,470.4^2}$$

$$F_{CD} = 4,475.5 \text{ lb}$$

THE CORRECT ANSWER IS: B

55. $r_{EW} = r_{NS}$ $K_{EW} > K_{NS}$ Buckling in east-west direction will control.

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

56. ACI 318-14, Tables 19.3.1.1, 19.3.2.1, and 19.3.3.1 give 4.5%. Neglect tolerance per problem statement.
 $\therefore$ minimum = 4.5%

THE CORRECT ANSWER IS: A

57. Concrete reaches ultimate strain and fails prior to steel yielding. Failure is sudden without warning (brittle failure).

THE CORRECT ANSWER IS: D

58. Douglas Fir-Larch No. 2, $F_c = 1,350$ psi Table 4A
 $F_c^* = (F_c)(C_D)(C_M)(C_t)(C_F)(C_i)$ Table 4.3.1
 $= (1,350)(1.6)(1.0)(1.0)(1.15)(1.0) = 2,484$ psi
 $F_c' = (F_c^*)(C_p)$ Table 4.3.1
 $= (2,484)(0.33) = 819.72$ psi
 $A = bd = (1.5)(3.5) = 5.25$ in²
 $P = F_c' A = (819.72)(5.25) = 4,303$ lb Table 4.3.1

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

59. Reference: TMS 402/602, Paragraph 8.3.5.1.4

The allowable shear stress resisted by the steel reinforcement is given by:

$$F_{vs} = 0.5 \left[\frac{A_v F_s d_v}{A_{nv} s} \right] \quad \text{Equation 8-30}$$

where

0.5 is an empirical factor

Commentary Section 9.3.4.1.2.2

A_v = cross-sectional area of shear reinforcement

F_s = 32,000 psi for Grade 60 reinforcement

Section 8.3.3.1 (b)

A_{nv} = net cross-sectional area of member

The allowable shear force V is given by :

$$V = F_{vs} A_{nv}$$

Rearranging Equation 2-30:

$$\begin{aligned} V &= 0.5 \left[\frac{A_v F_s d_v}{s} \right] \\ &= 0.5 \left(\frac{0.2(32,000)(24)}{8(1,000)} \right) \\ &= 0.5(19.2) \\ &= 10 \text{ kips} \end{aligned}$$

THE CORRECT ANSWER IS: A

60. Reference: ACI 318-14, Section 6.5, Table 6.5.2

Two or more spans not exceeding 10 ft.

Spans are equal.

Loads are uniformly distributed.

$LL \leq 3DL$

Prismatic members

$$l_n = 10 \text{ ft} - 1 \text{ ft} = 9 \text{ ft per Sec. 6.5}$$

$$W_u = 1.2 \times 100 + 1.6 \times 50 = 200 \text{ plf}$$

Use $W_u l_n^2 / 12$ for ALL supports.

Note: Beam/column stiffness not applicable to slabs

$$M_u = W_u l_n^2 / 12 = 1.35 \text{ ft-kips/ft}$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

- 61.**
- The properties of a W10×22 are as follows:

AISC, 14th ed., Table 3-2.

$$\phi_b M_{px} = 97.5 \text{ ft-kips}$$

$$M_{px}/\Omega_b = 64.9 \text{ ft-kips}$$

$$BF/\Omega_b = 2.68 \text{ kips (ASD) or } \phi_b BF = 4.02 \text{ kips (LRFD)}$$

$$L_p = 4.7 \text{ ft}$$

$$L_b = 10 \text{ ft}$$

$$\text{ASD: } M_{px}/\Omega_b - BF(L_b - L_p) = M_n/\Omega_b$$

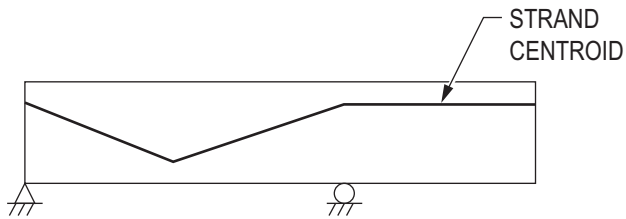
$$64.9 - 2.68(10 - 4.7) = 50.7 \text{ ft-kips}$$

$$\text{LRFD: } \phi_b M_{px} - BF(L_b - L_p) = \phi_b M_n$$

$$97.5 - 4.02(10 - 4.7) = 76.2 \text{ ft-kips}$$

THE CORRECT ANSWER IS: B

- 62.**

**THE CORRECT ANSWER IS: A**

PE CIVIL: STRUCTURAL SOLUTIONS

63. $\Delta_{\text{conc. load}} = \frac{PL^3}{28 EI}$ AISC, 14th ed., Beam Design

$$\Delta_{\text{uniform}} = \frac{5 WL^4}{384 EI} \Rightarrow \text{Set } \Delta_{\text{conc. load}} + \Delta_{\text{uniform}} = L/240$$

$$I \left(\frac{L}{240} \right) = \frac{PL^3}{28 E} + \frac{5 WL^4}{384 E}$$

$$\begin{aligned} I_{\text{required}} &= \frac{240}{L} \left[\frac{PL^3}{28 E} + \frac{5 WL^4}{384 E} \right] \\ &= \frac{240}{(15)(12)} \left[\frac{30(15 \times 12)^3}{28 (29,000)} + \frac{5 (1/12)(15 \times 12)^4}{384 (29,000)} \right] \\ &= 1.33(215.5 + 39.3) = 339 \text{ in}^4 \end{aligned}$$

- A. W12×40 $I = 307 \text{ in}^4$
- B. W12×45 $I = 348 \text{ in}^4 > 339$ OK
- C. W12×50 $I = 391 \text{ in}^4 > 339$ OK
- D. W14×26 $I = 245 \text{ in}^4$
- E. W14×30 $I = 291 \text{ in}^4$
- F. W14×34 $I = 340 \text{ in}^4 > 339$ OK
- G. W16×26 $I = 301 \text{ in}^4$
- H. W16×31 $I = 375 \text{ in}^4 > 339$ OK

THE CORRECT ANSWERS ARE: B, C, F, H

64. $A' = \text{Area of flange} = 6 \text{ in.}(0.5 \text{ in.}) = 3 \text{ in}^2$
 $\bar{y}' = \text{distance from neutral axis to centroid of flange} = 5.75 \text{ in.}$
 $Q_{\text{flange}} = 3 \text{ in}^2 (5.75 \text{ in.}) = 17.25 \text{ in}^3$
 $q = \frac{VQ}{I} = \frac{20 \text{ kips}(17.25 \text{ in}^3)}{232 \text{ in}^4} = 1.5 \text{ kips/in.}$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

65. Reaction at Joint A = $1/2$ total applied load (due to symmetry)
 $= (1/2)(50 + 100 + 50) = 100$ kips

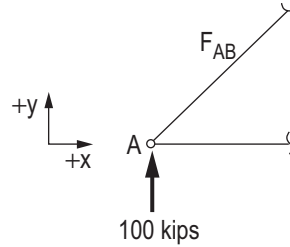
Considering Joint A:

$$\sum F_y = 0$$

$$+\frac{3}{5}F_{AB} + 100 = 0$$

$$F_{AB} = -\frac{5}{3}(100)$$

$$= -167 \text{ kips (compression)}$$



THE CORRECT ANSWER IS: D

66.
$$F_{cE} = \frac{0.822 E'_{\min}}{(l_e/d)^2}$$

NDS Section 3.7.1

$$E'_{\min} = E_{\min} \times C_M \times C_t \times C_i \times C_T$$

where:

$$E_{\min} = 440,000 \text{ psi (given)}$$

$$C_m = 1.0 \text{ (dry service)}$$

$$C_t = 1.0 \text{ (normal temperature)}$$

$$C_i = 1.0 \text{ (given)}$$

$$C_T = 1.0 \text{ (not a truss member)}$$

$$E'_{\min} = 440,000 \text{ psi} \times 1.0 \times 1.0 \times 1.0 \times 1.0$$

$$= 440,000 \text{ psi}$$

$$\text{Check } l_e/d: \quad 3 \text{ 1/2-in. direction} \quad 8 \times 12 / 3.5 = 27.5$$

$$1 \text{ 1/2-in. direction} \quad 4 \times 12 / 1.5 = 32 \quad \text{Controls}$$

$$F_{cE} = \frac{0.822 \times 440,000}{32^2} = 353 \text{ psi}$$

THE CORRECT ANSWER IS: B

67.
$$\frac{(1.6 \text{ kips})(3 \text{ in.})}{(8.25 \text{ in.} - 0.50 \text{ in.})} = 0.619 \text{ kips} = 619 \text{ lb}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

68. By inspection P controls.

$$M_n = M_p = F_y Z_x \leq 1.6 M_y$$

$$F_y = 36 \text{ ksi}$$

$$S_x = 1/6 bd^2 = 1/6 (6)(0.375)^2 = 0.141 \text{ in}^3$$

$$M_y = F_y S_x = (36)(0.141) = 5.076 \text{ in.-kips}$$

$$Z_x = 1/4 bd^2 = 1/4 (6)(0.375)^2 = 0.211 \text{ in}^3$$

$$M_p = F_y Z_x = (36)(0.211) = 7.6 \text{ in.-kips} \quad \leftarrow \text{governs}$$

$$\text{Check } M_p \leq 1.6 M_y \quad 1.6(5.076) = 8.1 > 7.6 \quad \therefore \text{OK}$$

AISC ASD:

$$M_n / \Omega = 7.6 / 1.67 = 4.55 \text{ in.-kips}$$

$$P_{\text{allow}} = \frac{4.55}{\left[4 - (1/2)(3/8)\right]} = 1.19 \sim 1.2$$

AISC LRFD:

$$\phi M_n = 0.9(7.6) = 6.84 \text{ in.-kips}$$

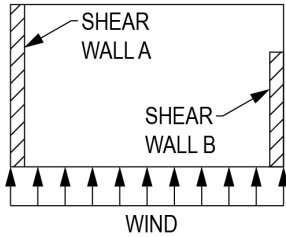
$$\phi P_n = \frac{6.84}{\left[4 - (1/2)(3/8)\right]} = 1.79 \sim 1.8$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

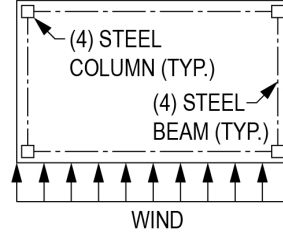
69. Reference: ASCE 7-10, 3rd printing, Section 12.3.1.3.

- WOOD
- DIAPHRAGM DEFLECTION CAN BE TOLERATED



Flexible

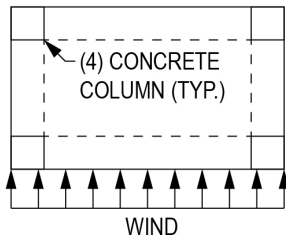
- METAL DECK SUPPORTED BY STEEL BEAMS AND COLUMNS



Flexible

Torsionally regular

- PT CONCRETE SLAB SUPPORTED BY CONCRETE COLUMNS/BEAMS

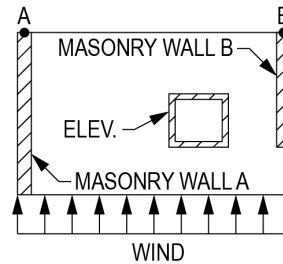


Rigid

Torsionally regular

- CONCRETE SLAB SUPPORTED BY MASONRY WALLS

$$A_{\text{STORYDRIFT}} = 1.5 \left(\frac{A_{\text{STORYDRIFT}} + B_{\text{STORYDRIFT}}}{2} \right)$$



Torsionally irregular

Rigid

THE CORRECTLY LABELED DIAGRAMS ARE SHOWN ABOVE.

PE CIVIL: STRUCTURAL SOLUTIONS

70. Overturning moment, $M_{ot} = 20 \text{ kips} \times 22 \text{ ft} = 440 \text{ ft-kips}$

$$\text{Restoring moment, } M_r = 80 \text{ kips} \times 2 \text{ ft} + \frac{2.5 \text{ kips/ft} \times (24 \text{ ft})^2}{2} = 880 \text{ ft-kips}$$

$$\text{Footing weight, } W_{fg} = 2.5 \text{ kips/ft} \times 24 \text{ ft} = 60 \text{ kips}$$

$$\text{Total vertical load, } P_{tot} = 80 \text{ kips} - 20 \text{ kips} + 60 \text{ kips} = 120 \text{ kips}$$

$$\text{Resultant location, } y = \frac{M_r - M_{ot}}{P_{tot}} = \frac{880 \text{ ft-kips} - 440 \text{ ft-kips}}{120 \text{ kips}} = 3.7 \text{ ft}$$

Since resultant is outside the middle third of the footing base:

$$P_{max} = \frac{2}{3} \times \frac{(P_{tot})}{(y \times \text{footing width})} = \frac{2 (120 \text{ kips})}{3 (3.7 \text{ ft} \times 8 \text{ ft})} = 2.7 \text{ ksf}$$

THE CORRECT ANSWER IS: D

71. $d \approx 24 \text{ in.} - (3 \text{ in.})_{cov} - (1 \text{ in.})_{\#8} = 20 \text{ in.}$ (given) Use d at $\bar{C}_L$ of #8 layers

b_o at $d/2$ from face (critical section)

$$b_o = 4(32 \text{ in.}) = 128 \text{ in.}$$

Per ACI 318-14, Section 22.6.5.2, Table 22.6.5.2, Section 22.6.5.3,

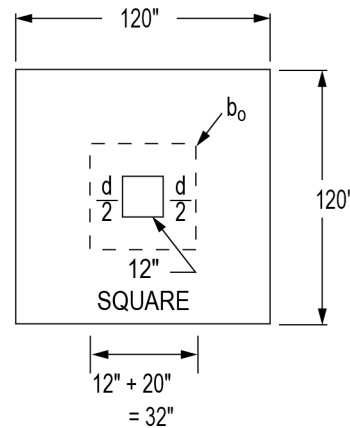
use smallest of the following:

a. $\left(2 + \frac{4}{1/1}\right) = 6$

b. $\left[\frac{(40)(20 \text{ in.})}{128 \text{ in.}} + 2\right] = 8.3$

c. 4 Controls

$$\begin{aligned} \phi V_n &= 0.75(4)(1)(\sqrt{3,000})(128 \text{ in.})(20 \text{ in.}) \\ &= 420.6 \text{ kips} \end{aligned}$$



THE CORRECT ANSWER IS: D

72. Critical case is skin friction resistance = applied uplift load.

$$\frac{(20 \text{ ft})\pi D(200 \text{ psf})}{1,000} = 100 \text{ kips}$$

$$D = \frac{100 \text{ kips} \times 1,000}{(200 \text{ psf} \times 20 \text{ ft} \times \pi)}$$

$$D = 7.96 \text{ ft}$$

THE CORRECT ANSWER IS: A

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73. AISC, 14th ed.

ASD:

$$F_{nv}/\Omega = 13.5 \text{ ksi}$$

$$\text{Allowable load} = 2(13)(0.79) = 21.2 \text{ kips}$$

LRFD:

$$\phi R_n = \phi F_{nv} A_b$$

$$\phi F_{nv} = 20.3 \text{ ksi (A307 bolts)}$$

$$\phi R_n = (20.3)(0.79)(2) = 31.9 \text{ kips}$$

Table 7-1

THE CORRECT ANSWER IS: B

74. Vertical reaction to Footing $B = 150 \text{ kips} \times 5 \text{ ft}/6 \text{ ft} = 125 \text{ kips}$

$$\text{Footing spring constant } K = 100 \text{ lb/in}^3 \times 24 \text{ in.} \times 24 \text{ in.} = 57,600 \text{ lb/in.}$$

$$F = K\Delta$$

$$\Delta = F/K = 125 \times 1,000/57,600 = 2.2 \text{ in.}$$

THE CORRECT ANSWER IS: A

75. Reference: PCI, Chapter 4.

$$\text{Total prestress force, } P = 2 \times 6 \text{ strands} \times 0.153 \text{ in}^2/\text{strand} \times 175 \text{ ksi} = 321.3 \text{ kips}$$

$$\text{Axial stress} = P/A = 321.3/(36 \times 18) = 0.496 \text{ ksi (compression)}$$

$$\text{Bending stress due to prestress} = Pe/S \quad S = bd^2/6 = 18 \times 36^2/6 = 3,888 \text{ in}^3$$

$$e = 36/2 - 3 - 2/2 = 14 \text{ in.}$$

$$\text{Bending stress due to prestress} = 321.3 \times 14/3,888 = 1.158 \text{ ksi (tension at top)}$$

$$\text{Bending stress due to beam self-weight} = 0.65 \text{ ksi (compression at top)}$$

$$\text{Thus, stress at top at release} = 0.49 - 1.15 + 0.65 = -0.01 \text{ ksi (tension)}$$

THE CORRECT ANSWER IS: A

76. Reference: ASCE 7-10, 3rd printing, Design Loads Table 3.2-1, p. 9.

$$85 \text{ lb/ft}^2$$

THE CORRECT ANSWER IS: C

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77. Design spectral response acceleration at short periods: IBC 2015, 1613.3.4

$$S_{DS} = \frac{2}{3} S_{MS} \quad (\text{Equation 16-39})$$

Adjusted maximum considered earthquake spectral response acceleration for short periods: IBC 2015, 1613.3.3

$$S_{MS} = F_a S_s \quad (\text{Equation 16-37})$$

For Site Class E and mapped short period maximum considered earthquake spectral acceleration: IBC 2015, Table 1613.3.3(1)

$$S_s = 100\% g$$

$$F_a = 0.9$$

$$\therefore S_{MS} = 0.9 \times 1.0 = 0.9$$

$$S_{DS} = \frac{2}{3} \times 0.9 = 0.6$$

THE CORRECT ANSWER IS: A

78. Reference: AASHTO Section 9.7.2.3.

$$\begin{aligned} L_{\text{eff}} &= 8.5 \text{ ft} - b_f + \text{flange overhang} \\ &= 8.5 \text{ ft} - 1.5 \text{ ft} + 0.25 \text{ ft} \\ &= 7.25 \text{ ft (7 ft 3 in.)} \end{aligned}$$

THE CORRECT ANSWER IS: C

79. $R = \left[(0.5' (150 \text{ pcf}) + 50 \text{ psf} + 12 \text{ psf}) + (0.83' (150 \text{ pcf}) + 50 + 12) \right] (6' \times 6')$
 $= 11,646 \text{ lb}$

THE CORRECT ANSWER IS: C

80. Reference: ACI 318-14.

ACI 318 26.12.3.1(b)

Concrete is unsatisfactory. Every arithmetic average of any three consecutive strength tests should be equal to or exceed the required f'_c .

$$\frac{4,215 + 4,160 + 3,625}{3} = 4,000; \quad \frac{4,160 + 3,625 + 4,010}{3} = 3,931; \quad \frac{3,625 + 4,010 + 4,015}{3} = 3,883$$

THE CORRECT ANSWER IS: A